

IDENTIFICATION

PRODUCT CODE: MAINDEC-08-DHMCA-A1-D
PRODUCT NAME: PDP8-E MEMORY EXTENSION
AND TIME SHARE CONTROL TEST
DATE CREATED: JUNE 16, 1972
MAINTAINER: DIAGNOSTIC PROGRAMMING GROUP
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M.C.O. DATE: JUNE 27, 1975

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MAIN DEC CHANGE NOTICE
MAY BE REQUIRED FOR
PROGRAM TO OPERATE

1. ABSTRACT

THIS PROGRAM TESTS THE MEMORY EXTENSION AND TIME SHARE CONTROL LOGIC FOR PROPER OPERATION, THE PROGRAM EXERCISES AND TESTS ALL IOT'S ASSOCIATED WITH MEMORY EXTENSION AND TIME SHARE CONTROL.

ERRORS ENCOUNTERED DURING RUNNING WILL RESULT IN A PROGRAM "HALT" OR A "JUMP TO SELF", WHICH MAY OCCUR IN ANY FIELD DEPENDING ON THE PORTION OF THE TEST EXECUTED. ERRORS MAY BE IDENTIFIED BY REFERENCING THE PROGRAM LISTING.

2. REQUIREMENTS

2.1 EQUIPMENT

PDP8-E COMPUTER WITH THE KMB-E OPTION INSTALLED AND AT LEAST 4K OF EXTENDED MEMORY.

2.2 STORAGE

THE PROGRAM REQUIRES 4200(8) LOCATIONS OF CORE MEMORY AND MUST RESIDE IN FIELD 0 ONLY.

2.3 PRELIMINARY PROGRAMS

ALL THE PROGRAMS FOR THE BASIC PDP8-E MUST HAVE BEEN RUN SUCCESSFULLY.

3. LOADING PROCEDURE

3.1 METHOD

THE PROGRAM IS LOADED INTO "FIELD 0" USING THE STANDARD BINARY LOADER TECHNIQUE.

4. STARTING PROCEDURE

4.1 CONTROL SWITCH SETTINGS

SR 9, 10, AND 11 MUST CONTAIN AN OCTAL VALUE EQUAL TO THE NUMBER OF EXTENDED FIELDS AVAILABLE. NOTE THAT FIELD 0 IS NOT INCLUDED.

SR0=0 WILL RESULT IN COMPLETE PROGRAM EXECUTION OF THE MEMORY

EXTENSION AND TIME SHARE CONTROL.

SR0=1 WILL LOOP THE PROGRAM ON THE MEMORY EXTENSION PORTION AND TEST THAT THE TIME SHARE IS DISABLED.

SR1=1 WILL RESULT IN AN END OF TEST HALT AT LOCATION 1565(8).

4.2 STARTING ADDRESS

THE STARTING ADDRESS IS LOCATION 0200(8).

4.3 OPERATOR ACTION

4.3.1 MEMORY EXTENSION AND TIME SHARE CONTROL (TIME SHARE ENABLED)

WITH THE PROGRAM IN MEMORY, SET THE SWITCH REGISTER TO 0000.

PRESS EXTENDED ADDRESS LOAD.

SET THE REGISTER TO 0200 OCTAL.

PRESS ADDRESS LOAD.

PLACE THE OCTAL VALUE OF EXTENDED FIELDS AVAILABLE IN SR9-11.

PRESS CLEAR AND THEN CONTINUE.

THE PROGRAM SHOULD RUN UNTIL A FAILURE OCCURS OR UNTIL STOPPED BY THE OPERATOR WITH SR1=1. NOTE THAT THE PROGRAM SHOULD ALWAYS BE STOPPED WITH SR1=1.

THE TTY BELL WILL SIGNAL A SUCCESSFUL TEST AT THE COMPLETION OF EVERY PASS.

4.3.2 MEMORY EXTENSION PORTION (TIME SHARE DISABLED)

WITH THE PROGRAM IN MEMORY, SET THE SWITCH REGISTER TO 0000.

PRESS EXTENDED ADDRESS LOAD.

SET THE SWITCH REGISTER TO 0200 OCTAL.

PRESS ADDRESS LOAD.

PLACE THE OCTAL VALUE OF EXTENDED FIELDS AVAILABLE IN SR9-11.

PLACE SR0=1 TO EXECUTE MEMORY EXTENSION ONLY.

PRESS CLEAR AND THEN CONTINUE.

THE PROGRAM SHOULD HALT AT LOCATION 3651(8), THIS WILL

VERIFY THAT THE TIME SHARE IS DISABLED. ALL OTHER ERRORS
AT THIS TIME WILL BE CONSIDERED AS AN ILLEGAL CONDITION.

PRESS CONTINUE.

THE PROGRAM SHOULD LOOP UNTIL AN ERROR OCCURS OR UNTIL STOPPED
BY THE OPERATOR WITH SR1=1.

THE TTY BELL WILL SIGNAL A SUCCESSFUL TEST AT THE END
OF EVERY PASS.

5. OPERATING PROCEDURE -----

5.1 OPERATOR ACTION -----

5.1.1 MEMORY EXTENSION AND TIME SHARE CONTROL -----7-----

VISUALLY VERIFY THAT THE TIME SHARE DISABLE JUMPER IS "OUT"
ON THE M837 MODULE AND FOLLOW THE OPERATOR ACTION IN 4.3.

5.1.2 MEMORY EXTENSION PORTION -----

VISUALLY VERIFY THAT THE TIME SHARE DISABLE JUMPER IS "IN"
ON THE M837 MODULE AND FOLLOW THE OPERATOR ACTION 4.3.

6. ERRORS -----

6.1 ERROR DESCRIPTION -----

BOTH "HALTS" AND "JUMP TO SELF" ARE USED TO INDICATE ERROR
CONDITIONS. IN EITHER CASE REFER TO THE PROGRAM LISTING
FOR MORE INFORMATION.

6.2 ERROR RECOVERY -----

ALL ERRORS ENCOUNTERED MUST BE CORRECTED BEFORE PROCEEDING
ON IN THE PROGRAM.

7. RESTRICTIONS -----

7.1 OPERATING RESTRICTIONS -----

PDP8-E ONLY WITH THE KM8-E OPTION INSTALLED AND AT LEAST 4K
OF EXTENDED MEMORY.

THE NUMBER OF EXTENDED AVAILABLE FIELDS MUST BE IN SR9-11.

IF MEMORY EXTENSION ONLY, THE TIME SHARE MUST BE DISABLED AND SR0=1.

IF MEMORY EXTENSION AND TIME SHARE CONTROL, THE TIME SHARE MUST BE ENABLED AND SR0=0.

IN ALL CASES SR1=1 MUST BE USED TO STOP PROGRAM.

THE PROGRAM MUST RESIDE IN FIELD 0 ONLY.

BOTH PORTIONS OF THE TEST MUST BE RUN, 4.3.1 AND 4.3.2, TO VERIFY THAT THE TIME SHARE CAN BE DISABLED AND ENABLED.

8. MISCELLANEOUS -----

8.1 EXECUTION TIME -----

EXECUTION TIME DEPENDS ON THE AMOUNT OF AVAILABLE EXTENDED FIELDS. EXECUTION TIME FOR 32K APPROXIMATIVELY 3.75 MINUTES.

9. PROGRAM DESCRIPTION -----

THE PROGRAM EXERCISES AND TESTS ALL IOT'S ASSOCIATED WITH THE MEMORY EXTENSION AND TIME SHARE CONTROL; THE ABILITY TO RUN WITH THE TIME SHARE DISABLED; THE ABILITY TO RUN "EXECUTIVE" AND "USER MODES" IN ALL AVAILABLE FIELDS WITH THE TIME SHARE ENABLED; THE ABILITY TO REFERENCE ALL MEMORY FIELDS FROM FIELD 0 AND VICE-VERSA; THE ABILITY TO READ AND WRITE DATA IN ALL AVAILABLE FIELDS AND THE ABILITY TO RUN PROGRAM INTERRUPTS AND INTERRUPT INHIBIT IN ALL FIELDS.

THE TIME SHARE OPTION DEVELOPES A NEW MODE OF OPERATION OR THE "USER MODE". ALL HLT, OSR, AND IOT INSTRUCTIONS ARE ILLEGAL IN USER MODE AND SHOULD "TRAP OUT". THE PROGRAM WILL THEN DETERMINE IF AN ERROR CONDITION DOES EXIST. IN SOME CASES, IN TIME SHARING, AN ERROR CONDITION CANNOT BE INDICATED WITH A "HLT" OR "TYPE OUT" BECAUSE THIS WOULD BE ILLEGAL. THEREFORE A "JUMP TO SELF" IS USED TO INDICATE ERRORS.

9.1 TEST 00 -----

TEST CDF AND RDF FOR ALL COMBINATIONS 0 TO 7.

9.2 TEST 01 -----

TEST INTERRUPT BUFFER BITS 9-11 WITH RIB. PI IS ENABLED AND TTY FLAG IS USED FOR INTERRUPTS. DO ALL COMBINATIONS 0 TO 7.

9.3 TEST 02

TEST DCA I AND TAD I TO ALL AVAILABLE FIELDS. EACH STACK
WILL CONTAIN ITS DF# IN LOCATION 7000.

9.4 TEST 03

TEST CIF INSTRUCTION. CHECKS THE ABILITY OF A CIF-ION-
NOP-JMP AND CIF-ION-NOP-JMS.

9.5 TEST 04

TEST GTF INSTRUCTION FOR ITT FLAG AND SAVE FIELD.
GET SAVE FIELD AFTER INTERRUPT AND CHECK INTERRUPT
INHIBIT. DO ALL COMBINATIONS 0 TO 7.

9.6 TEST 05

TEST ION AND LINK FROM RTF. TEST INTERRUPT INHIBIT BEFORE
PI. GET THE FLAGS WITH GTF.

9.7 TEST 06

TEST READ AND WRITE DATA IN ALL AVAILABLE EXTENDED FIELDS.

9.8 TEST 07

CONFIDENCE CHECK ON ALL EXISTENT FIELDS. MAKE SURE ALL
STACKS ARE ACCESSED CORRECTLY.

9.9 TEST 08

TEST DF AND IF FROM SAVE FIELD AFTER PI. USE RTF TO
SET THE FLAGS AND GTF TO GET THE FLAGS. CHECK INTERRUPT
INHIBIT. DO ALL SF COMBINATIONS 0 TO 77.

9.10 TEST 09

TEST PROGRAM INTERRUPT IN ALL AVAILABLE EXTENDED FIELDS.
USE RTF, GTF, RDF, AND RIF FOR CHECK.

9.11 TEST 10

TEST INTERRUPT INHIBIT IN ALL AVAILABLE EXTENDED FIELDS.
TEST CIF-ION-JMP COMBINATION.

9.12 TEST 11

TEST SAVE FIELD WITH RMF IOT.

9.13 TEST 12

TEST AUTO-INDEX IN ALL AVAILABLE EXTENDED FIELDS.

9.14 TEST 13

DYNAMIC RMF TEST. TEST ALL SF TO DE TRANSFERS AND SF
TO IB TRANSFERS.

9.15 TEST 14

TEST NON-EXISTENT FIELDS FOR ALL DIS. IF 32K PRESENT
BY-PASS TEST.

9.16 TEST 15

TEST TIME SHARE IN FIELD 0.

9.17 TEST 16

TEST TIME SHARE IN ALL AVAILABLE EXTENDED FIELDS.

10. LISTING

/PDP8-E, MEMORY EXTENSION AND TIME SHARE CONTROL TEST.

/

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/STARTING ADDRESS IS 0200.

/CONSTANTS

/

0201	CDF=6201
0202	CIF=6202
0214	RDF=6214
0224	RIF=6224
0244	RHF=6244
0234	RIB=6234
0274	SUF=6274
0264	CUF=6264
0254	SINT=6254
0204	CINT=6204
0007	CAF=6007
0005	RTF=6005
0004	GYF=6004
0001	ION=6001
0002	IOF=6002
0000	SKON=6000
0003	SRQ=6003
0040	SPF=6040
0041	TSF=6041
0032	KCC=6032
0002	IOF=6002
0036	KRB=6036
0000	IOT=6000
7421	HQL=7421
	/
0000	*0
0000	0000
0001	5001
0002	0002
0003	0003
	/
0020	*20
	/
0020	JMPI0, JMP I 0
0021	IS00, IS0 0
0022	XTFLG, TFLG
0023	XSTKS, NSTKS
0024	XRMF, TRMF
0025	XTRANS, TRANS
0026	XAUTO, TAUTO
0027	LOOP, 0
0030	NDF, 0
0031	STKS, 0
0032	DAT, 0
0033	NOSTAK, 0
0034	NOFLD, 0
0035	KCAIM, CAI-1

0036	1133	KCAI,	CAI
0037	7402	KHLT,	HLT
0040	6201	KCDF,	6201
0041	6202	KCIF,	6202
0042	1316	XFD,	EXFD
0043	0001	K1,	1
0044	0007	K7,	7
0045	0010	K10,	10
0046	7777	K7777,	7777
0047	7000	K7000,	7000
0050	7707	K7707,	7707
0051	7767	K7767,	7767
0052	7757	K7757,	7757
0053	7747	K7747,	7747
0054	7737	K7737,	7737
0055	7727	K7727,	7727
0056	7717	K7717,	7717
0057	7776	K7776,	7776
0060	7775	K7775,	7775
0061	7774	K7774,	7774
0062	7773	K7773,	7773
0063	7772	K7772,	7772
0064	7771	K7771,	7771
0065	0067	POINT,	.*2
0066	0067	K7S,	.*1
0067	7766	K7766,	7766
0070	7755		7755
0071	7744	K7744,	7744
0072	7733		7733
0073	7722		7722
0074	7711		7711
0075	7700		7700
0076	1127	XTDF,	STDF
0077	1130	XTDF1,	STDF+1
0100	1302	KXFLD,	EXFLD
0101	5402	KJMP,	JMP I 2
0102	1200	KNTR,	ENTER
0103	0020	K20,	20
0104	5505	JMP2,	JMP I KFLD0
0105	1427	KFLD0,	RTRN
0106	1422	KRTN,	CA0+2
0107	1400	XFIB,	SFIB
0110	7770	K7770,	7770
0111	0070	K0070,	0070
0112	0000	XSAV,	0000
0113	7770	XCOUNT,	7770
0114	0000	XTOR,	0000
0115	5200	K5200,	5200
0116	1200	K1200,	1200
0117	0077	K0077,	0077
0120	0011	K0011,	0011
0121	7700	K7700,	7700
0122	0002	K0002,	0002

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0123 0004 K0004, 0004
0124 7402 K7402, 7402
0125 6000 K6000, 6000
0126 0100 K0100, 0100
0127 0203 PLACE, BEGIN
0130 1000 K1000, 1000
0131 2600 TIME, T1
0132 0017 K0017, 0017
0133 6001 K6001, 6001
0134 5535 JMPIR, JMP I XRET
0135 2511 XRET, RET
0136 0000 XDATA, 0000
0137 0000 K0000, 0000
0140 0003 K0003, 0003
0141 0001 K0001, 0001
0142 1100 K1100, 1100
0143 7745 SROO, 7745
0144 3577 K3577, 3577
0145 7745 K7745, 7745
0146 3633 XSR0, XSR0
0147 1556 XELL, BELL*1
0150 1555 XBELL, BELL
0151 6046 TTB, TLS
0152 3643 XTRAP, TRAP
0153 5531 ATRAP, JMP I TIME
0154 0000 FCO, 0000
0155 2047 XDATER, DATER
0156 6211 KCDF1, CDF 10
0157 2525 KDATER, 2525

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/TEST 00
/TEST CDF AND RDF, USE CDF TO SET THE DATA
/FIELD AND RDF TO READ THE DATA FIELD,
/DO ALL COMBINATIONS 0 TO 7,
/

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0200 0200
0200 7604 BEGIN1, LAB
0201 7510 SPA
0202 5552 JMP I XTRAP
0203 7300 BEGIN, CLA CLL
0204 6007 CAP
0205 6264 CUP
0206 1037 TAD KHLT /STORE A HLT IN LOC. 1 AND
0207 3001 DCA 1 /CHECK FOR STRAY INTERRUPT RQST.
0210 6001 ION
/
0211 6201 DFB, CDF 00 /DF 0
0212 6214 RDF
0213 7450 SNA /SHOULD NOT SKIP
0214 5220 JMP DF7
0215 7402 HLT /ERROR, CDF OR RDF FAILED

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0216 7200 CLA
0217 5211 JMP DF0 /REPEAT
/
0220 1050 DF7, TAD K7707 /7707
0221 6271 CDF 70 /DF 7
0222 6214 RDF
0223 7040 CMA /AC = 0
0224 7450 SNA /SHOULD NOT SKIP
0225 5231 JMP OK1
0226 7402 HLT /CDF OR RDF FAILED
0227 7200 CLA
0230 5220 JMP DF7
/
0231 2027 OK1, ISB LOOP /CHECK DONE
0232 5211 JMP DF0
/
0233 7200 CLA
0234 3027 DCA LOOP /LOOP COUNTER
/
0235 1051 DF1, TAD K7767 /7767
0236 6211 CDF 10 /DF 10
0237 6214 RDF
0240 7040 CMA /AC=0
0241 7450 SNA
0242 5246 JMP DF2
0243 7402 HLT /CDF 1 OR RDF FAILED
0244 7200 CLA
0245 5235 JMP DF1
/
0246 1052 DF2, TAD K7757 /7757
0247 6221 CDF 20 /DF2
0250 6214 RDF
0251 7040 CMA /AC=0
0252 7450 SNA
0253 5257 JMP OK2
/
0254 7402 HLT /CDF 2 OR RDF FAILED
0255 7200 CLA
0256 5246 JMP DF2
/
0257 2027 OK2, ISB LOOP /DONE IF SKP
0260 5235 JMP DF1
0261 7200 CLA
0262 3027 DCA LOOP
/
0263 1053 DF3, TAD K7747 /7747
0264 6231 CDF 30 /DF 3
0265 6214 RDF
0266 7040 CMA /AC=0
0267 7450 SNA
0270 5274 JMP DF4
0271 7402 HLT /CDF 3 OR RDF FAILED
0272 7200 CLA
0273 5263 JMP DF3

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0274 1854 / DF4, TAD K7737 /7737
0275 6241 CDF 48 /DF 4
0276 6214 RDP
0277 7848 CMA /AC=8
0300 7458 SNA
0301 5385 JMP OK3
0302 7482 HLT /CDF 4 OR RDP FAILED
0303 7288 CLA
0304 5274 JMP DF4

0305 2827 / OK3, ISE LOOP /DONE IF SKP
0306 5263 JMP DF3

0307 7288 / CLA
0310 3827 DCA LOOP

0311 1855 / DF5, TAD K7727 /7727
0312 6251 CDF 58 /DF5
0313 6214 RDP
0314 7848 CMA /AC=8
0315 7458 SNA
0316 5322 JMP DF6
0317 7482 HLT /CDF 5 OR RDP FAILED,
0320 7288 CLA
0321 5311 JMP DF5

0322 1856 / DF6, TAD K7717 /7717
0323 6241 CDF 68 /DF 6
0324 6214 RDP
0325 7848 CMA /AC=8
0326 7458 SNA
0327 5333 JMP OK4

0330 7482 HLT /CDF 6 OR RDP FAILED
0331 7288 CLA
0332 5322 JMP DF6

0333 2827 / OK4, ISE LOOP /DONE IF SKP
0334 5311 JMP DF5
0335 6888 SKON /SKP IF ION
0336 7482 HLT /IS ION STILL ON

/TEST 81
/NOW TEST INTERRUPT BUFFER (IB) BITS 9-11 WITH
/RIB, PI IS ENABLED. TELEPRINTER FLAG IS
/USED FOR INTERRUPT. DO ALL COMBINATIONS 8 TO 7.

0337 6281 / CDF 88 /DF8
0340 1828 TAD JMP18 /JMP18=JMP I 8
0341 3881 DCA 1 /C(1)=JMP I 8
0342 3827 DCA LOOP
0343 6841 TSP /TEST TTY FLAG
0344 4422 JMS I XTFLG /SET FLAG

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0345 6881 IB8, ION /ENABLE PI
0346 7288 CLA
0347 6234 RIB /READ SF
0350 7458 SNA
0351 5354 JMP IB1
0352 7482 HLT /RIB FAILED
0353 5345 JMP IB8

0354 6211 / IB1, CDF 18 /DF 1
0355 6881 ION
0356 7288 CLA
0357 6214 RDP /DF SHOULD BE 8 AFTER A PI
0360 7458 SNA
0361 5364 JMP ,+3
0362 7482 HLT
0363 5354 JMP IB1 /DF NOT CLEARED, OR NO PI

0364 1857 / TAD K7776
0365 6234 RIB /READ SF
0366 7848 CMA /AC=8
0367 7458 SNA
0370 5373 JMP OK5
0371 7482 HLT /RIB OR SF FAILED
0372 5354 JMP IB1
0373 2827 OK5, ISE LOOP /DONE IF SKP
0374 5345 JMP IB8
0375 5776 JMP I ,+1
0376 8488 IB2-2

0400 8488 +488
0400 7288 CLA
0401 3827 DCA LOOP

0402 6221 / IB2, CDF 28 /DF 2
0403 6881 ION
0404 7288 CLA
0405 6214 RDP /SHOULD BE 8 AFTER PI
0406 7458 SNA
0407 5212 JMP ,+3
0410 7482 HLT /DF NOT CLEARED, OR NO PI
0411 5282 JMP IB2

0412 1868 / TAD K7775
0413 6234 RIB /AC=7777
0414 7848 CMA /=8
0415 7458 SNA
0416 5221 JMP IB3
0417 7482 HLT /RIB OR SF FAILED
0420 5282 JMP IB2

0421 6231 / IB3, CDF 38 /DF3
0422 6881 ION
0423 7288 CLA
0424 6214 RDP /DF SHOULD BE CLEARED

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0425 7450 SNA
0426 5231 JMP ,+3
0427 7402 HLT /DF NOT CLEARED
0430 5221 JMP IB3

0431 1061 TAD K7774
0432 6234 RIB /AC=7777
0433 7040 CMA /AC=0
0434 7450 SNA
0435 5240 JMP OK6
0436 7402 HLT /RIB OR SF FAILED
0437 5221 JMP IB3

0440 2027 OK6, ISL LOOP /DONE IF SKP
0441 5202 JMP IB2

0442 7200 CLA
0443 3027 DCA LOOP

0444 6241 IB4, CDF 40 /DF 3
0445 6001 ION
0446 7200 CLA
0447 6214 RDF /DF MUST BE 000 AFTER A PI
0450 7450 SNA /ERROR IF SKIP
0451 5254 JMP ,+3

0452 7402 HLT /DF NOT 0 AFTER PI
0453 5244 JMP IB4

0454 1062 TAD K7773 /AC=7773
0455 6234 RIB /AC=7777
0456 7040 CMA /AC=0
0457 7450 SNA
0460 5263 JMP IB5
0461 7402 HLT /RIB OR SF FAILED
0462 5244 JMP IB4

0463 6251 IB5, CDF 50 /DF5
0464 6001 ION
0465 7200 CLA
0466 6214 RDF /DF SHOULD=000
0467 7450 SNA
0470 5273 JMP ,+3
0471 7402 HLT /DF NOT 0 AFTER PI
0472 5263 JMP IB5

0473 1063 TAD K7772 /AC = 7772
0474 6234 RIB /AC = 7777
0475 7040 CMA /AC = 0000
0476 7450 SNA
0477 5302 JMP OK7
0500 7402 HLT /RIB OR SF FAILED
0501 5263 JMP IB5

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0502 2027 OK7, ISL LOOP /DONE IF 0 AND SKIP
0503 5244 JMP IB4

0504 7200 CLA
0505 3027 DCA LOOP

0506 6261 IB6, CDF 60 /DF6
0507 6001 ION
0510 7200 CLA
0511 6214 RDF /DF MUST=0 AFTER PI
0512 7450 SNA
0513 5316 JMP ,+3
0514 7402 HLT /DF NOT 0 AFTER PI
0515 5306 JMP IB6

0516 1064 TAD K7771 /7771
0517 6234 RIB /AC=7777
0520 7040 CMA
0521 7450 SNA
0522 5325 JMP IB7
0523 7402 HLT /RIB OR SF FAILED
0524 5306 JMP IB6

0525 6271 IB7, CDF 70 /DF 7
0526 6001 ION
0527 7200 CLA
0530 6214 RDF /DF MUST = 0 AFTER PI
0531 7450 SNA
0532 5335 JMP ,+3
0533 7402 HLT /DF NOT 0
0534 5325 JMP IB7

0535 1110 TAD K7770 /AC=7777
0536 6234 RIB
0537 7040 CMA
0540 7450 SNA
0541 5344 JMP OK8
0542 7402 HLT /RIB OR SF FAILED
0543 5325 JMP IB7

0544 2027 OK8, ISL LOOP /DONE IF SKP
0545 5306 JMP IB6
0546 5747 JMP I ,+1 /NEW PAGE
0547 0600 600

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0600 *600
/TEST 02
/NOW TEST DCA I AND TAD I TO ALL STACKS, NUMBER OF
/EXTENDED STACKS SHOULD BE IN 000 TO 11. EACH STACK WILL
/CONTAIN ITS DF# IN LOCATION 7000.
0600 3027 DCA LOOP

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0601 4423 DCAI, JMS I XSTKS /READ SR 9-11
0602 7001 IAC
0603 3030 DCA NDF /DF NUMBER = 1 TO START
0604 1040 TAD KCDF /6201
0605 1045 TAD K10
0606 3207 DCA ,+1 /DF 001 TO START WITH
0607 6201 DFLD, CDF 00 /WILL BE INCREMENTED
0610 1030 TAD NDF /DF#
0611 3447 DCA I K7000 /PUT IN 7000 OF STACK
0612 2031 ISB STKS /ALL STACKS WHEN 0
0613 7410 SKP
0614 5222 JMP TADI /TEST TAD I
0615 1045 TAD K10
0616 1207 TAD DFLD /INCR. CDF 101
0617 3207 DCA DFLD
0620 2030 ISB NDF
0621 5207 JMP DFLD

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0622 4423 TADI, JMS I XSTKS /SR0-11 AGAIN
0623 7001 IAC
0624 3030 DCA NDF /DF#+1 AGAIN
0625 1040 TAD KCDF /6201
0626 1045 TAD K10
0627 3230 DCA ,+1
0630 6201 TFLD, CDF 00 /AC=DF CONTENTS NOW
0631 1447 TAD I K7000 /SAVE TEMP
0632 3032 DCA DAT
0633 1032 TAD DAT
0634 7041 CIA /2'S COMP
0635 1030 TAD NDF /BETTER BE EQUAL
0636 7040 SEA GLA
0637 5252 JMP CAA-1 /ERROR PATH
0640 2031 ISB STKS /ALL WHEN 0
0641 5245 JMP ,+4
0642 2027 ISB LOOP /DONE WHEN 0
0643 5201 JMP DCAI
0644 5256 JMP ISF /NEXT TEST
0645 1045 TAD K10
0646 1230 TAD TFLD /CDF 101 + 10
0647 3230 DCA TFLD
0650 2030 ISB NDF
0651 5230 JMP TFLD

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0652 1032 / TAD DAT /DATA AS READ
0653 7402 CAA, HLT /AC=DATA READ
0654 7200 CLA
0655 5230 JMP TFLD

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/TEST B3

/CIF TEST, CHECKS THE ABILITY OF A CIF-ION-NOP-JMP OR
 /CIF=ION-NOP-JMS SEQUENCE TO DO THE FOLLOWING:
 /1. CIF ENABLE MB TO IB TRANSFER,
 /2. INHIBIT INTERRUPT TILL JMP OR JMS EXECUTED,
 /3. INTERRUPT AFTER JMP OR JMS EXECUTED,

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0656 6201 /SET UP FOR CIF-ION-NOP-JMP CHECK.
IBSF, CDF 00 /SET LOCS 1-2 TO ISB 0,
0657 1021 TAD ISB0 /JMP I 0 RESPECTIVELY.
0660 3001 DCA 1
0661 1352 TAD KNOP
0662 3002 DCA 2
0663 1020 TAD JMP10
0664 3003 DCA 3

/
/NOW STORE HALTS IN LOC1, CIFJMP+1,
/AND CIFJMS+1 OF ALL EXTENDED FIELDS.

0665 4423 JMS I XSTKS
0666 1040 TAD KCDF
0667 1045 TAD K10
0670 3271 DCA ,+1
0671 0211 HLTS, CDF 10
0672 1037 TAD KHLT
0673 3443 DCA I K1
0674 1037 TAD KHLT
0675 3754 DCA I CAB
0676 1037 TAD KHLT
0677 3755 DCA I CAG
0700 2031 ISB STKS
0701 7410 SKP
0702 5305 JMP ,+3
0703 1271 TAD HLTS
0704 5267 JMP HLTS-2
0705 6201 CDF 00
0706 0041 TSF /ENSURE T10 FLAG SET,
0707 4422 JMS I XTCLG /SET COUNTER FOR 4096 PASSES.
0710 3027 DCA LOOP /INITIALIZE TO CIF 00.
0711 1041 AGAIN1, TAD KCIF
0712 3323 DCA CIFJMP
0713 3353 DCA CIFCK /INITIALIZE I.F. CHECK TO 0,
0714 4423 JMS I XSTKS /READ SR9-11,
0715 1323 CIFJPL, TAD CIFJMP
0716 1045 TAD K10
0717 3323 DCA CIFJMP
0720 1353 TAD CIFCK
0721 1045 TAD K10
0722 3353 DCA CIFCK
0723 6202 CIFJMP, CIF 00 /MODIFIED TO CURRENT FIELD
/UNDER TEST.

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0724 6001 ION
0725 7000 NOP
0726 5327 JMP ,+1 /ERROR. NO PI OR INHIBIT PI.
0727 7402 HLT
0730 6234 RIB
0731 7041 CIA

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0732 1353      TAD      CIECK
0733 7650      SNA CLA
0734 9344      JMP      CAD+3
0735 1353      TAD      CIECK
0736 7421      MQL      /LOAD MQ
0737 7300      CLA CLL
0740 6234      RIB
0741 7402      HLT
0742 7200      CAD,    CLA
0743 5323      JMP      CIEJMP
                     /ERROR. I.B. TO I.F. TRANSFER
                     /FAILED AFTER CIE-JMP. BAD
                     /I.F. IN AC, GOOD I.F. IN
                     /MQ. REPEAT UPON CONTINUE.
                     /DONE?
0744 2031      ISB      STKS
0745 5315      JMP      CIEJPL
0746 2027      ISB      LOOP
0747 5311      JMP      AGAIN1
0750 5751      JMP I    .+1
0751 1000      ISBF1
0752 7000      KNOP,    NOP
0753 0000      CIECK,   0
0754 0724      CAB,    CIEJMP+1
0755 1020      CAC,    CIEJMS+1
                     /NO. DO NEXT FIELD
                     /4096 TIMES?
                     /NO. DO IT ALL AGAIN.
                     /YES. GO TEST CIE-JMS,

1000 1000      *1000
1000 7200      ISBF1,   CLA
1001 6201      CDF      00
1002 6041      TSF
1003 4422      JMS I    XTFLG
1004 3027      DCA      LOOP
1005 1041      AGAIN2,  TAD      KCIF
1006 3217      DCA      CIEJMS
1007 3246      DCA      CIECK1
1010 4423      JMS I    XSTKS
1011 1217      CIEJSL,  TAD      CIEJMS
1012 1045      TAD      K10
1013 3217      DCA      CIEJMS
1014 1246      TAD      CIECK1
1015 1045      TAD      K10
1016 3246      DCA      CIECK1
1017 6202      CIEJMS,  CIE      00
                     /MODIFIED TO CURRENT FIELD
                     /UNDER TEST.

1020 6001      ION
1021 7000      NOP
1022 4223      JMS      .+1
1023 0000      0
1024 7402      HLT
1025 6234      RIB
1026 7041      CIA
1027 1246      TAD
1030 7650      SNA CLA
1031 5241      JMP      CAD+3
1032 1246      TAD      CIECK1
1033 7110      CLL RAR
1034 7012      RTR

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1035 6234      RIB
1036 7402      CAE,    HLT
1037 7200      CLA
1040 5217      JMP      CIEJMS
                     /ERROR. I.B. TO I.F. TRANSFER
                     /FAILED AFTER CIE-JMS. BAD
                     /I.F. IN AC=8, GOOD I.F.
                     /IN AC=11. REPEAT UPON CONTINUE
                     /DONE?
1041 2031      ISB      STKS
1042 5211      JMP      CIEJSL
1043 2027      ISB      LOOP
1044 5205      JMP      AGAIN2
1045 5647      JMP I    XGTF1
                     /NO. DO NEXT FIELD.
                     /4096 TIMES?
                     /NO. DO IT ALL AGAIN.
                     /YES. GO ON TO NEXT TEST

1046 0000      CIECK1,0
1047 2271      XGTF1,  GTF1

/
/TEST 10
/TEST INTERRUPT INHIBIT
/FROM EACH FIELD. REFER TO HEADING TITLED "EXTENDED
/FIELD TEST ROUTINE". THIS ROUTINE IS PLACED IN
/EACH TESTED FIELD AT THE ADDRESSES SPECIFIED. THE
/INDICATED ERROR HALTS WILL BE IN THE EXTENDED
/FIELD. PRESS CONT. TO RECOVER. ONLY 1 FIELD WILL
/CONTAIN THE ROUTINE AT ANY ONE TIME. OTHER FIELDS
/WILL CONTAIN ALLOIS. THE ROUTINE IS REPLACED WITH
/HALTS AFTER COMPLETION. THE PORTIONS OF THE FIELD
/WHICH DO NOT CONTAIN THE ROUTINE ARE SET TO 0000
/BEFOREHAND.
/
/
/SETUP FIELDS TO TEST, POINTERS, ETC.,
/
1050 4423      TRMF,   JMS I    XSTKS      /READ SR9-11
1051 1040      TAD      KCDF      /6201
1052 3260      DCA      .+6
1053 1260      TAD      .+5
1054 1045      TAD      K10
1055 3260      DCA      .+3
1056 7040      CMA
1057 3010      DCA 10
1060 6201      CDF 00
1061 3410      DCA I 10
1062 1010      TAD 10
1063 7040      CMA
1064 7640      SZA CLA
1065 5261      JMP ,+4
1066 2031      ISB STKS
1067 5253      JMP TRMF+3
                     /PLACE 0'S IN EACH FIELD FROM
                     /LOC. 0 TO 777.

/
/NOW PUT A HLT IN EACH FIELD IN THE SAME
/LOCATION AS CAL. BELOW.
/
1070 4423      JMS I    XSTKS      /READ SR 9-11
1071 1040      TAD      KCDF
1072 1045      TAD      K10

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1073 3274      DCA ,+1
1074 0201      CDF 00
1075 1036      TAD KCAI      /KCAI = ADDRESS OF CAI,
1076 3027      DCA LOOP      /SAVE TEMPORARILY
1077 1037      TAD KHLT      /KHLT = 7402 (HLT)
1100 3427      DCA I LOOP
1101 2031      ISB STKS      /DONE ALL STACKS WHEN SKIP
1102 7410      SKP
1103 5306      JMP ,+3
1104 1274      TAD CHDF
1105 5272      JMP CHDF-2
/
1106 0201      CDF 00
1107 0041      STRMF,      /CHECK TTY FLAG
1110 4422      JMS I XTFLG      /GO SET IT
1111 1050      TAD K7707
1112 3027      DCA LOOP
1113 1065      TAD POINT
1114 3066      DCA K7S      /POINTER FOR K7700 TO K7706
1115 4423      JMS I XSTKS      /READ SR 9-11
1116 1040      TAD KCDF      /0201
1117 1045      TAD K10      /10
1120 3327      DCA STDF
1121 1041      TAD KCIF      /0202
1122 1045      TAD K10      /10
1123 3330      DCA STDF+1
1124 1330      TAD STDF+1
1125 3442      DCA I XFD
1126 4425      JMS I XTRANS      /PUT TEST ROUTINE INTO FIELD X
/
1127 0211      STDF,      CDF 10      /FIELD 1 TO START WITH
1130 0212      CIF 10
1131 5332      JMP ,+1      /SHOULD ENTER EXTENDED FIELD
                              /AFTER THIS JMP, HLT IF NOT
1132 7000      NOP
1133 7402      CAI,      HLT      /ERROR, PI FAILED
                              /C(AC) = C(I.0.)
1134 5327      JMP STDF      /REPEAT SAME TEST.
/
/ENTER HERE AFTER PI FROM EXTENDED BANK
1200 01200
1200 0214      ENTER,      RDF      /DF SHOULD BE 000
1201 7450      SNA      /ERROR IF SKIP
1202 5206      JMP ,+4      /CHECK C(SF)
1203 7402      HLT      /AC=C(DF)
1204 7200      CLA
1205 5476      JMP I XTDF      /REPEAT TEST
1206 0212      CIF 10      /SET I.0. TO FIELD 1
1207 0244      RMF      /I.0. NOW EQUAL TO SF
1210 0234      RIB      /READ 10
1211 0202      CIF 00
1212 0201      CDF 00

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1213 1466      TAD I K7S
1214 7040      CMA
1215 7650      SNA CLA      /ERROR IF SKIP
1216 5226      JMP OKPC
1217 0244      RMF
1220 0234      RIB
1221 7402      HLT      /ERROR RMF AND PI WORKED, BUT
                              /I.0. NOT CORRECT AFTER RMF,
                              /AC=C(I0)
1222 7200      CLA
1223 0201      CDF 00
1224 0202      CIF 00
1225 5476      JMP I XTDF      /BACKUP A PAGE AND REPEAT
/
1226 1036      CKPC,      TAD KCAI      /KCAI=ADDRESS OF CAI
1227 7001      IAC      /MAKE CAI+1
1230 7041      CIA
1231 1000      TAD 0      /COMPARE TO C(0)
1232 7650      SNA CLA      /SHOULD NOT SKIP
1233 5240      JMP ,+5      /ALL OK SETUP FOR NEXT FIELD
1234 1000      TAD 0
1235 7402      HLT      /ERROR, ALL WORKED, BUT
                              /C(IPC) WAS NOT=TO CAI+1
                              /AFTER PI IN EXTENDED
                              /FIELD, C(AC)=C(0),F0.
                              /CHECK FOR PI NOT INHIBITED,
                              /OR AUTO-INDEX REG.
                              /12 FAILING IN THE EXTENDED FIELD.
1236 7200      CLA
1237 5476      JMP I XTDF      /BACKUP AND REPEAT
/
/SETUP FOR NEXT FIELD
/
1240 2031      ISB STKS      /DONE ALL IF SKIP
1241 5246      JMP ,+5
1242 2027      ISB LOOP      /DONE LOOPING IF SKIP
1243 5645      JMP I ,+2      /REPEAT ALL AGAIN
1244 5507      JMP I XFI0
1245 1113      STRMF+4      /BACK TO LAST PAGE
/
/SET LAST TESTED FIELD TO ALL 0'S AND PUT A
/HLT IN RESPECTIVE ADDRESS OF CAI
/
1246 7240      CLA CMA
1247 3010      DCA 10
1250 1476      TAD I XTDF      /CDF X0 AT STDF
1251 3252      DCA ,+1
1252 0211      CDF 10      /F1 TO START WITH
1253 3410      DCA I 10
1254 1010      TAD 10
1255 7040      CMA
1256 7640      SNA CLA      /CLEAR0 IF SKIP
1257 5253      JMP ,+4
1260 0201      CDF 00

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1261 1476      TAD I XTDF      /CDF XB AT STOF
1262 3263      DCA ,*1
1263 6211      CDF 10
1264 1037      TAD KHLT      /*7402 (HLT)
1265 3436      DCA I KCAI      /KCAI=ADDRESS OF CAI
1266 6201      CDF 00      /RESTORE DF
/
/INCREMENT CDF AND CIF 10'S AT STOF, STDF+1
/TO NEXT FIELD.
/
1267 1476      TAD I XTDF      /CDF XB AT STOF
1270 1045      TAD K10
1271 3476      DCA I XTDF
1272 1477      TAD I XTDF1      /CIF XB AT STOF
1273 1045      TAD K10
1274 3477      DCA I XTDF1
1275 1477      TAD I XTDF1
1276 3316      DCA EXFD
1277 2066      ISZ K7S
1300 4321      JMS TRANS      /PUT ROUTINE IN NEW FIELD
1301 3476      JMP I XTDF      /TEST NEW FIELD
/
/EXTENDED FIELD TEST ROUTINE
/
/THE FOLLOWING INSTRUCTIONS ARE PLACED IN
/EACH EXTENDED FIELD TESTED. THE NUMBERS IN THE
/COMMENTS FIELD CORRESPOND TO THE
/MEMORY LOCATIONS IN THE TESTED FIELD. LOCATIONS
/0 THRU 11 ARE USED FOR AN ERROR ROUTINE
/IN CASE FIELD 0 IS NOT ENTERED AFTER AN
/INTERRUPT, THE EXTENDED FIELD SHOULD BE
/ENTERED AT LOCATION CAI-1 WHICH CORRESPONDS
/TO CAI-1 IN FIELD 0.
/
/EXTENDED FIELD INSTRUCTIONS:
/
1302 0000      EXFLD, 0      /0
1303 1000      TAD 0      /1
1304 7450      SNA      /IF LOC. 0 NOT *0 PI DIDN'T
/ENTER FIELD 0
1305 5010      JMP 10      /3
1306 7402      HLT      /4. INTERRUPTED TO THIS FIELD
/INSTEAD OF FIELD 0.C(AC)=C(0)
/WHICH SHOULD BE CAI-1
/IF NOT, CHECK LOC. 7777.IT
/MUST = 3412 (JMP I 12).
/
1307 7200      CLA      /5
1310 3000      DCA 0      /6
1311 5420      JMP I 20      /7. C(20) *CAI
1312 7402      HLT      /10. THE JMP I 12 AT LOC.
/7777 WAS NOT EXECUTED,
/OR INTERRUPT FAILED. IF
/NO INTERRUPT, LOCATION 12
/NOW CONTAINS 0 INSTEAD

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1313 5005      JMP 5      /OF ADDRESS CAI.
1314 1133      CAI      /11. REPEAT IN THIS FIELD
/12. AUTO-INDEX TO CAI+1
/IN P 0 IF THE JMP I 12
/WORKS.
/LOCS, 13 TO 17 ARE ALL 0'S
/
1315 1133      CAI      /20. EQUALS CAI IN F0.
/
/LOCS, 21 TO CAI-2 ARE ALL 0'S
/
1316 6212      EXFD, CIF 10 /FIELD 1 TO START WITH
1317 6001      ION      /LOC. CAI. SEE SYMBOL TABLE
/FOR CAI.
/LOCS, CAI+1 TO 7776 ARE ALL 0'S
/
1320 5412      JMP I 12      /7777. PI SHOULD OCCUR,
/AFTR THIS INSTRUCTION,
/TO FIELD 0.
/
/ROUTINE TO TRANSFER TEST ROUTINE TO PROPER FIELD
/
1321 0000      TRANS, 0
1322 1101      TAD KJMP      /KJMP=JMP I 2
1323 3001      DCA 1      /IN FIELD 0
1324 1102      TAD KNTR      /KNTR = LOC. ENTER
1325 3002      DCA 2      /OF FIELD 0
1326 1100      TAD KXFLD      /KXFLD = LOC. EXFLD
1327 3010      DCA 10
1328 3011      DCA 11
1329 1067      TAD K7766      /1-10 DECIMAL
1330 3000      DCA 0      /SAVE
1331 1476      TAD I XTDF      /CDF XB IN STOF
1332 3337      DCA ,*3
1333 6201      CDF 00
1334 1410      TAD I 10
1335 6211      TRFLD, CDF 10 /F1 TO START WITH
1336 3411      DCA I 11 /PUT IN EXTENDED FIELD
1337 2000      ISZ 0      /DONE LOCS 1 TO 12 IF SKIP
1338 5335      JMP ,*5
1339 1337      TAD TRFLD
1340 3347      DCA ,*3
1341 6201      CDF 00
1342 1410      TAD I 10
1343 6211      CDF 10
1344 3503      DCA I K20      /PUT E40 IN LOC. 20
1345 6201      CDF 00
1346 1337      TAD TRFLD
1347 3355      DCA ,*2
1348 1410      TAD I 10
1349 6211      CDF 10
1350 3435      DCA I KCAIM      /PUT CIF XB IN CAI-1
1351 6201      CDF 00

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1360 1337      TAD TRFLD
1361 3363      DCA ,+2
1362 1410      TAD I 10
1363 0211      CDF 10
1364 3436      DCA I KCA1      /ION TO LOC. CA1
1365 0201      CDF 00
1366 1337      TAD TRFLD
1367 3371      DCA ,+2
1370 1410      TAD I 10
1371 0211      CDF 10
1372 3446      DCA I K7777      /PUT JMP I 12 IN 7777
1373 0201      CDF 00
1374 5721      JMP I TRANS      /EXIT

1400          *1400
/
/TEST 11
/TEST SF WITH AN RMF IOT, AN INTERRUPT IN FIELD 0 IS CREATED, AFTER
/WHICH, THE OF AND IB REGISTERS ARE SET TO FIELD 1.
/THE SF SHOULD CONTAIN FIELD 0. THE TEST
/THEN MAKES SURE THE IB IS CLEARED, THEN SET BY ISSUING AN RMF.
/FOLLOWED BY A JMP I K7000, IF THE IB IS CLEARED, THE JMP GOES TO 7000 IN FIELD 0,
/IF THE IB AND SF ARE INCLUSIVE OR'D, THE JMP GOES TO 7000 IN FIELD 1, AND
/A HALT OCCURS THERE, RESTART FROM 1400 AFTER AN ERROR, THE TEST IS LOOPEO
/512 TIMES.

1400 0041      SFIB,      TSF          /SEE IF FLAG IS SET.
1401 4422      JMS I XTFLG      /SET IT
1402 1047      TAD K7000      /7000
1403 3027      DCA LOOP
1404 0211      CDF 10          /DF=FIELD 1
1405 1037      TAD KHLT        /HLT
1406 3447      DCA I K7000      /7000, FIELD 1=HLT
1407 0201      CDF 00          /DF=0
1408 1104      TAD JMP2        /JMP2=JMP I KFLD0
1409 1104      DCA I K7000      /7000, FIELD 0=JMP I KFLD0
1410 3447      /KFLD0=LOC. RTN
1411          /KJMP=JMP I 2

1412 1101      TAD KJMP
1413 3001      DCA 1
1414 1106      TAD KRTN        /KRTN=LOC. CAG+2
1415 3002      DCA 2

/
/BEGIN TEST
/
1416 0001      ION            /ENABLE PI
1417 7000      NOP
1420 7402      CAG,      HLT      /ERROR NO PI
1421 5200      JMP SFIB        /REPEAT TEST

/
/RETURN HERE AFTER PI
/
1422 7200      CLA
1423 0211      CDF 10          /DF=FIELD1
1424 0212      CIF 10          /IB=FIELD1
1425 0244      RMF            /IB SHOULD=FIELD0

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1426 5447      JMP I K7000      /IF SHOULD=FIELD0

/
1427 2027      RTRN,      ISE LOOP /WORKED OK
1430 5216      JMP CAG-2      /LOOP
1431 5232      JMP TAUTO      /DONE, GO TO NEXT TEST

/
/TEST 12
/TEST ALL AUTO-INDEX REGISTERS IN EACH EXTENDED FIELD.
/IDENTICAL TEST ROUTINES ARE PERFORMED FROM EACH FIELD,
/AND ERROR HALTS OCCUR IN THE FIELD CURRENTLY RUNNING
/THE ROUTINE, PRESS CONT, TO RESUME TESTING, EACH
/FIELD CONTAINS ALL 0'S EXCEPT FOR THE AREA OCCUPIED
/BY THE TEST ROUTINE, FIELD 0 IS RE-ENTERED
/AFTER EACH TEST, AND THE NEXT SEQUENTIAL FIELD
/IS THEN ENTERED, REFER TO THE HEADING "AUTO-
/INDEX TEST" FOR THE SEQUENCE OF OPERATIONS.

1432 0201      TAUTO,      CDF 00
1433 1050      TAD K7707
1434 3027      DCA LOOP        /LOOP COUNTER
1435 4423      JMS I XSTKS      /READ SR 9-11
1436 1040      TAD KCDF        /0201
1437 3246      DCA OFN
1440 1246      NEWDF,      TAD OFN
1441 1045      TAD K10          /INCREMENT OF
1442 3246      DCA OFN

/
/CLEAR ONE FIELD TO 0
/
1443 7040      CMA
1444 3010      DCA 10
1445 3000      DCA 0          /USE LOC. 0 FOR A COUNTER
1446 0211      CDF 10          /FIELD 1 TO START WITH
1447 3410      DCA I 10
1450 2000      ISE 0
1451 5247      JMP ,+2
1452 0201      CDF 00

/
/NOW PUT TEST ROUTINE IN THE EXTENDED FIELD
/
1453 1317      TAD D0AUTO      /1ST LOC. OF ROUTINE MINUS 1
1454 3010      DCA 10          /SOURCE
1455 1071      TAD K7744      /-20 DECIMAL
1456 3000      DCA 0          /USE LOC. 0 AS COUNTER
1457 1317      TAD D0AUTO
1458 3011      DCA 11          /DESTINATION
1459 1246      TAD OFN
1462 3265      DCA ,+3
1463 0201      CDF 00
1464 1410      MOVE,      TAD I 10
1465 0211      CDF 10          /FIELD 1 TO START
1466 3411      DCA I 11

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1467 2808      ISB 0      /MOVE WHEN SKIP
1478 5263      JMP MOVE

/
/NOH SET AUTO-I REGS 10 TO 17 TO 7777.
/
1471 1118      TAD K7770      /-8 DECIMAL
1472 3808      DCA 0
1473 1844      TAD K7      /7
1474 3818      DCA 10
1475 7848      CMA      /7777
1476 3418      DCA I 10
1477 2808      ISB 0      /10 TO 17 = 7777 WHEN SKIP
1500 5275      JMP ,=3
1501 7848      CMA
1502 3446      DCA I K7777      /PUT 7777 IN LOC. 7777 OF EXTENDED FIELD
1503 6214      RDP      /READ D.F.
1504 1841      TAD KCIF      /6202
1505 3306      DCA ,=1
1506 6212      CIP 10      /FIELD 1 TO START
1507 4716      JMS I FILDX      /ENTER EXTENDED FIELD
                                /510 OCTAL LOGS, BEFORE THE
                                /TAD I 10 INSTRUCTION.
                                /THIS IS A TEST OF THE
                                /DEFER BIT, 900 US DELAY

/
/ENTER FIELD 0 FROM EXTENDED FIELD HERE.
/
1510 2831      GOTO0, ISB 8TKS      /DONE ALL WHEN SKIP
1511 5240      JMP NEWDF      /SETUP FOR NEXT
1512 2827      ISB LOOP      /ALL DONE IF SKIP
1513 5235      JMP NEWDF-3      /REPEAT ALL
1514 5715      JMP I LBTP
1515 1600      LBTP, RMFTST
/
1516 1002      FILDX, DOAUTO-515

/
/      AUTO-INDEX TEST
/
/THE ROUTINE WILL BE PLACED IN THE SAME RESPECTIVE
/LOCATIONS IN EACH EXTENDED FIELD. ANY ERROR
//HALTS WILL OCCUR IN THE EXTENDED FIELD. PRESS
/CONTINUE TO PROCEED WITH TESTING. THE INDEX
/REGISTERS 10 TO 17 INITIALLY CONTAIN 7777, AND
/ARE AUTO-INDEXED TO 0000 BY A TAD I INSTRUCTION.
/A HALT OCCURS IF THE REG. IS NOT INCREMENTED TO 0.
/THE TAD I WOULD HAVE THEN REFERENCED LOC. 7777,
/WHICH CONTAINS 7777.
/

1517 1517      DOAUTO, .      /THIS LOC. IS NOT MOVED TO
                                /THE EXTENDED FIELD.

1520 7200      CLA

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1521 1410      TAD I 10
1522 7440      SZA
1523 7402      HLT      /ERROR, INDEX REG. 10 FAILED
1524 1411      TAD I 11
1525 7440      SZA
1526 7402      HLT      /INDEX REG. 11 FAILED
1527 1412      TAD I 12
1528 7440      SZA
1529 7402      HLT      /12 FAILED
1530 1413      TAD I 13
1531 7440      SZA
1532 7402      HLT      /13 FAILED
1533 1414      TAD I 14
1534 7440      SZA
1535 7402      HLT      /14 FAILED
1536 1415      TAD I 15
1537 7440      SZA
1538 7402      HLT      /15 FAILED
1539 1416      TAD I 16
1540 7440      SZA
1541 7402      HLT      /16 FAILED
1542 1417      TAD I 17
1543 7440      SZA
1544 7402      HLT      /17 FAILED
1545 6201      CDF 00      /SET UP TO FIELD 0
1546 6202      CIP 00      /SET I.B. TO FIELD 0
1547 5310      JMP GOTO0      /EXIT TO FIELD 0
/END OF TEST ROUTINE
/
/
/
/RING BELL AT THE COMPLETION OF TEST
/CHECK SR1=1 FOR HLT AT END OF TEST
/
1554 0007      AND 7
1555 1354      BELL, TAD ,=1
1556 6046      TLS      /RING BELL
1557 6041      TSP
1558 5357      JMP ,=1
1559 7404      LAB
1560 7004      RAL
1561 7500      SMA
1562 5527      JMP I PLACE      /START TEST OVER
1563 7402      HLT      /END OF TEST
1564 5527      JMP I PLACE      /HIT CONTINUE TO START TEST OVER
/
/TEST 13
/DYNAMIC RMF TEST.
/TESTS ALL SF TO DF TRANSFERS AND THOSE SF TO IB TRANSFERS
/AS APPLICABLE TO THE NUMBER OF EXTENDED FIELDS PRESENT.
/THE GENERAL METHOD IS TO INTERRUPT FROM EACH EXTENDED FIELD
/WITH THE DF=FROM 0 THROUGH 7, AN RMF INSTRUCTION IS THEN ISSUED
/AND CONTROL TRANSFERRED TO AN EXTENDED FIELD. THE RMFDY ROUTINE

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    /IN THAT FIELD THEN CHECKS THAT THE IF AND DF ARE CORRECT, IF NOT,
    /THE FAILING IF OR DF IS IN THE IF OR DF REG, AND THE CORRECT FIELD
    /NUMBER IS IN AC BITS 6 THRU 8.
    /
    /
    1600 *1600
    /
    1600 7604 RMFTST, LAS /CHECK HOW MANY EXTENDED FIELDS
    1601 0044 AND K7 /ARE PRESENT
    1602 7041 CIA /NEGATE AND SAVE.
    1603 3205 DCA IFGN
    1604 4700 JMS I XFERP /TRANSFER RMFDY ROUTINE TO ALL
    1605 0000 IFGN, 0 /EXTENDED FIELDS.
    1606 7744 -34
    1607 1702 RMFDY-1
    1608 3275 DCA LBSTSTC /SET RMFTST COUNTER FOR 4096 PASSES
    1609 1302 TAD JMP14 /SET INTERRUPT LINK.
    1610 3001 DCA 1
    1611 1274 TAD INIEP
    1612 3004 DCA 4
    1613 6201 RMFL3, CDF 00 /INITIALIZE IF TO 0.
    1614 3341 DCA KIFSHB
    1615 1205 TAD IFGN /INITIALIZE TEST COUNTER
    1616 3276 DCA RMFCN1
    1617 1341 RMFL2, TAD KIFSHB /UPDATE CURRENT IF.
    1618 1045 TAD K10
    1619 3341 DCA KIFSHB
    1620 1341 TAD KIFSHB
    1621 7041 CIA
    1622 3342 DCA MIFSHB
    1623 1110 TAD K7770 /INITIALIZE DF COUNTER TO -10.
    1624 3277 DCA DFON
    1625 1110 TAD K7770 /INITIALIZE DF TO -10.
    1626 3337 DCA KDFSHB
    1627 1337 RMFL1, TAD KDFSHB /UPDATE DF.
    1628 1045 TAD K10
    1629 3337 DCA KDFSHB
    1630 1337 TAD KDFSHB
    1631 7041 CIA
    1632 3340 DCA MDFSHB
    1633 1205 TAD IFGN /TRANSFER OF AND IF INFORMATION
    1634 3244 DCA .02 /TO EXTENDED FIELDS.
    1635 4700 JMS I XFERP
    1636 0000 0
    1637 7774 -4
    1638 1736 KDFSHB-1
    1639 6201 CDF 00
    /
    1650 1040 TAD KCDF /UPDATE CDF INST.
    1651 1337 TAD KDFSHB
    1652 3260 DCA RMFI1
    1653 1041 TAD KCIF /UPDAT CIF INST.
  
```

```

    1654 1341 TAD KIFSHB
    1655 3261 DCA RMFI2
    1656 0041 RMFE2, TSF
    1657 4422 JMS I XTELG /ENSURE TIO FLAG SET.
    1658 6201 RMFI1, CDF /SET OF AND IF TO CURRENT FIELD.
    1659 6202 RMFI2, CIF
    1660 5303 JMP RMFDY /GO TO RMFDY IN CURRENT IF.
    1661 6244 INTE, RMF /ENTER FROM INTERRUPT FROM EX. FLD.
    1662 5310 JMP RMFDY1 /GO BACK TO EXTENDED FIELD.
    1663 2277 RMFE1, ISB /ALL OF'S USED WITH CURRENT IF.
    1664 5233 JMP RMFL1 /NO. DO NEXT DF.
    1665 2276 ISB RMFCN1 /ONE PASS OF RMFTST COMPLETE?
    1666 5221 JMP RMFL2 /NO. DO NEXT IF.
    1667 2275 ISB LBSTSTC /RMFTST DONE?
    1668 5215 JMP RMFL3 /NO. DO AGAIN.
    1669 9701 JMP I XHEM /YES. GO TO NEXT TEST
    1670 1663 INTEP, INTE
    1671 0000 LBSTSTC, 0
    1672 0000 RMFCN1, 0
    1673 0000 DFON, 0
    1674 2000 XFERP, XFER
    1675 2200 XHEM, NOHEM
    1676 5404 JMP14, JMP I 4
    /
    /
    /
    /ROUTINE TO CHECK CORRECT TRANSFERS FOR SAVE FIELD TO DATA FIELD AND
    /SAVE FIELD TO INST, BUFFER TO INSTRUCTION FIELD AFTER
    /RMF,
    /STORED IN ALL EXTENDED FIELDS.
    /
    1703 0001 RMFDY, ION /THIS IS NOT TRANSFERRED.
    1704 7000 NOP
    1705 0002 IOF
    1706 7402 HLT /INTERRUPT FAILURE.
    1707 5333 JMP REPEAT
    1708 7200 RMFDY1, CLA
    1709 6214 ROP /CHECK FOR CORRECT DATA FIELD
    1710 1340 TAD MDFSHB
    1711 7650 SNA CLA
    1712 5320 JMP .04
    1713 1337 TAD KDFSHB /DATA FIELD INCORRECT
    1714 7402 HLT /SF TO DF TRANSFER FAILED AFTER RMF.
    1715 5333 JMP REPEAT /REPEAT THIS TEST.
    1716 6224 RIF /CHECK FOR CORRECT INSTRUCTION FIELD.
    1717 1342 TAD MIFSHB
    1718 7650 SNA CLA
    1719 5327 JMP .04
    1720 1341 TAD KIFSHB /INSTRUCTION FIELD INCORRECT.
    1721 7402 HLT /SF TO IB TRANSFER FAILED AFTER RMF
    1722 5333 JMP REPEAT /REPEAT THIS TEST.
  
```

```

1727 6201 CDF 00 /GO BACK AND RUN NEXT TEST.
1730 6202 CIF 00
1731 5732 JMP I .+1
1732 1665 RMFE1
1733 6201 REPEAT, CDF 00 /GO BACK AND REPEAT FAILING
1734 6202 CIF 00 /TEST.
1735 5736 JMP I .+1
1736 1656 RMFE2
1737 0000 KDFSHB, 0 /DATA FIELD SHOULD BE
1740 0000 MDFSHB, 0 /TWO'S COMPLEMENT OF ABOVE.
1741 0000 KIFSHB, 0 /INSTRUCTION FIELD SHOULD BE
1742 0000 MIFSHB, 0 /TWO'S COMPLEMENT OF ABOVE
/
/
/
/
/ROUTINE TO TRANSFER N1 WORDS STARTING AT P IN FIELD 0 TO P IN THE
/NEXT N2 EXTENDED FIELDS.
/THE CALLING SEQUENCE IS:
/JMS I XFERP
/-N2
/-N1
/P=1

2000 /
*2000 /
2000 0000 XFER, 0
2001 7200 CLA
2002 1600 TAD I XFER /GET -N2
2003 3242 DCA N2
2004 2200 ISZ XFER /GET -N1
2005 1600 TAD I XFER
2006 3243 DCA N1
2007 2200 ISZ XFER /GET P-1
2008 1600 TAD I XFER
2009 3244 DCA P
2010 2200 ISZ XFER /UPDATE TO RETURN ADDRESS,
2011 1040 TAD KCDF /INITIALIZE CDF INST,
2012 3232 DCA XFERIN
2013 1242 TAD N2
2014 3245 DCA XFERC2
2015 1244 XFERL2, TAD P /PUT POINTER IN AUTO 10 AND 11.
2016 3010 DCA 10
2017 1244 TAD P
2018 3011 DCA 11
2019 1243 TAD N1 /SET COUNTER 1 TO -N1
2020 3246 DCA XFERC1
2021 1232 TAD XFERIN /UPDATE CDF INST,
2022 1045 TAD K10
2023 3232 DCA XFERIN

```

```

2030 6201 XFERL1, CDF 00 /TRANSFER
2031 1410 TAD I 10
2032 6201 XFERIN, CDF 00
2033 3411 DCA I 11
2034 2246 ISZ XFERC1 /DONE WITH CURRENT FIELD?
2035 5230 JMP XFERL1 /NO, CONTINUE.
2036 2245 ISZ XFERC2 /DONE WITH ALL FIELDS?
2037 5217 JMP XFERL2 /NO, DO NEXT FIELD
2040 6201 CDF 00 /ALL DONE, SET DF#0.
2041 5600 JMP I XFER /EXIT.
2042 0000 N2, 0
2043 0000 N1, 0
2044 0000 P, 0
2045 0000 XFERC2, 0
2046 0000 XFERC1, 0
/
/TEST 06
/NO DO A READ AND WRITE DATA TEST IN
/ALL AVAILABLE EXTENDED FIELDS.
/IF A FAILURE OCCURS CHECK LOC, 10
/FOR BAD ADDRESS ABFA AND LOC, RANA
/FOR THE MOST RECENT FIELD CHANGE.
/LOC, KOATER CONTAINS DATA PATTERN USED.
/
2047 0000 DATER, 0000
2050 7300 CLA CLL
2051 4423 JMS I XSTKS
2052 1040 TAD KCDF
2053 1045 TAD K10
2054 3257 DCA RANA /MODIFIED UNDER TEST
2055 7340 CLA CLL CMA
2056 3010 DCA 10 /SET AUTO REGISTER
2057 6201 RANA, CDF
2060 4276 JMS FILL /LOAD UP FIELD WITH DATA
2061 7340 CLA CMA CLL
2062 3010 DCA 10
2063 4312 JMS CHECK /CHECK DATA IN FIELD
2064 7300 CLA CLL
2065 2031 ISZ STKS
2066 7410 SKP
2067 5274 JMP .+5
2070 1257 TAD RANA
2071 1045 TAD K10
2072 3257 DCA RANA /CHECK NEXT FIELD
2073 5255 JMP RANA -2
2074 6201 CDF
2075 5647 JMP I DATER
/
/ROUTINE TO FILL FIELD WITH DATA
/
2076 0000 FILL, 0000
2077 7300 CLA CLL
2100 1137 TAD KOATER
2101 3410 DCA I 10
2102 1137 TAD KOATER

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```

2103 7040      CMA
2104 3410      DCA I 10
2105 1010      TAD 10
2106 7001      IAC
2107 7640      SZA CLA
2110 5277      JMP FILL +1
2111 5676      JMP I FILL

/ROUTINE TO CHECK DATA IN FIELD
/
2112 0000      CHECK, 0000
2113 7300      CLA CLL
2114 1410      TAD I 10
2115 7001      IAC
2116 1410      TAD I 10
2117 7440      SZA          /AC CONTAINS BAD BITS
2120 7402      HLT          /MEMORY CONTROL WORKED BUT
2121 7300      CLA CLL      /DATA PATTERN FAILURE IN
2122 1010      TAD 10      /EXTENDED MEMORY.
2123 7001      IAC
2124 7640      SZA CLA      /IS CHECK DONE
2125 5313      JMP CHECK +1
2126 5712      JMP I CHECK

/
/

2200          +2200
/
/TEST 14
/REFERENCE ALL 4K FIELDS NOT PRESENT.
/IF 32K IS PRESENT, THE TEST IS BY-PASSED.
/EACH FIELD NOT PRESENT IS REFERENCED
/BY THE PROGRAM WITH JMP, DCA AND TAD.
/THE PROGRAM MUST CONTINUE IN SEQUENCE
/BELL WILL SIGNAL A SUCCESSFUL TEST
/
2200 7200      NOMEM, CLA
2201 1110      TAD K7770
2202 3027      DCA LOOP      /TEST LOOP COUNTER
2203 7604      LAS          /READ SR9-11
2204 0044      AND K7
2205 7041      CIA
2206 1044      TAD K7        /SUBTRACT MAX. POSSIBLE
2207 7450      SNA
2210 5546      JMP I XXSR#    /32K PRESENT, CAN'T TEST
2211 3033      DCA NOSTAK    /SAVE NO. MISSING
2212 3547      DCA I XELL    /CLEAR THE TLS IOT AT
                          /BELL+1 TO PROHIBIT
                          /FALSE INDICATION, TLS
                          /IS RESTORED LATER WRONG
                          /ENTRY FROM NON-EXISTENT

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```

/MEMORY MAY CAUSE A
/HANG-UP AT BELL+2 AND +3.
/# OF FIELDS PRESENT
2213 7604      LAS
2214 0044      AND K7
2215 7001      IAC          /+1 TO GET 1ST MISSING
2216 7100      CLL
2217 7006      RTL          /POSITION TO AC 6-8.
2220 7004      RAL
2221 3034      DCA NOFLD    /1ST MISSING
2222 1033      TAD NOSTAK   /# STACKS NOT HERE
2223 7041      CIA
2224 3033      DCA NOSTAK   /USED AS COUNTER

/
/
2225 1040      TAD K0F0     /6201
2226 1034      TAD NOFLD    /MISSING STACK
2227 3245      DCA CDF0S

/
/NOW READ ALL 0'S FROM ALL NON-EXISTENT FIELDS
/IF CONTROL PORTION ONLY, RING BELL.
/IF NOT PROCEED TO TIME SHARE.
/
2230 4244      JMS ALL0     /READ ALL 0 FROM 1ST
2231 2033      CNSTK, 1SE NOSTAK /DONE ALL MISSING IF SKIP
2232 5237      JMP POS
2233 2027      1SE LOOP     /DONE LOOPING IF SKIP
2234 5636      JMP I XNOM   /REPEAT
2235 5546      JMP I XXSR#

/
/
2236 2203      XNOM, NOMEM+3
/
/
2237 1245      POS, TAD CDF0S /OF PLUS 1
2240 1045      TAD K10
2241 3245      DCA CDF0S
2242 4244      JMS ALL0     /READ ALL 0'S
2243 5231      JMP CNSTK    /CHECK DONE

/
/ROUTINE TO READ ALL 0'S:
/
2244 0000      ALL0, 0
2245 0201      CDF0S, CDF 00 /SET DF TO 1ST MISSING
2246 7240      CLA CMA
2247 3010      DCA 10       /10 AND 11 USED FOR ADDRESS
2250 7040      CMA
2251 3011      DCA 11
2252 3002      DCA 2        /USE AS COUNTER
2253 7040      CMA
2254 3410      DCA I 10     /WRITE 1'S INTO NON-EXIS-
                          /TENT FIELD.
2255 2002      1SE 2

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2256 5253      JMP ,+3
2257 1411      TAD I 11      /READ NON-EXIST, FIELD
2260 7650      SNA CLA      /SHOULD = 0000
2261 5264      JMP ,+3
2262 1011      TAD I 11
2263 7402      CAX, HLT      /ERROR, AN EXISTING FIELD
                               /WAS REFERENCED, C(AC)=
                               /ADDRESS REFERENCED

2264 2002      ISB 2
2265 5257      JMP CAX-4      /READ NEXT

2266 6201      DONE0, CDF 00
2267 6202      CIP 00
2270 5644      JMP I ALL0      /EXIT

/
/
/TEST 04

/TEST GTF FOR FLAG AND SAVE FIELDS
/GET SAVE FIELDS AFTER INTERRUPT
/CHECK INTERRUPT INHIBIT, DO ALL
/COMBINATIONS 0 TO 7,
/

2271 7300      GTF1, CLA CLL
2272 1020      TAD JMP10      /SET FOR RETURN
2273 3001      DCA 1
2274 1040      TAD KCDF
2275 3304      DCA XSDF
2276 1304      MGTF, TAD XSDF      /GET FIRST FIELD
2277 0111      AND K0070
2300 7120      STL
2301 7010      RAR
2302 7012      RTR
2303 3112      DCA XSAV
2304 6201      XSDF, CDF 00
2305 6041      TSF      /IS TTY FLAG SET
2306 4422      JMS I XTFLG      /GET THE FLAG
2307 6001      ION
2310 7340      CLA CLL CMA      /CHECK FOR JAM ON GTF
2311 6004      GTF      /GET THE FLAG
2312 7041      CIA
2313 1112      TAD XSAV      /TTY + CURRENT FIELD
2314 7440      SEA
2315 7402      HLT      /FLAG + FIELD
2316 2027      ISB LOOP      /4096 TIMES
2317 5276      JMP MGTF
2320 1045      TAD K10
2321 1304      TAD XSDF
2322 3304      DCA XSDF
2323 2113      ISB XCOUNT      /MORE FIELDS TO CHECK
2324 5276      JMP MGTF
2325 1110      TAD K7770
2326 3113      DCA XCOUNT
2327 5730      JMP I XION1      /YES, GO TO NEXT TEST

```

```

2330 2331      XION1, ION1
/
/TEST 05
/TEST ION AND LINK FROM RTF
/TEST INTERRUPT INHIBIT BEFORE PI
/GET THE FLAG WITH GTF.
/

2331 7300      ION1, CLA CLL
2332 1021      TAD ISZ0
2333 3001      DCA 1
2334 1020      TAD JMP10
2335 3002      DCA 2
2336 6005      RTF
2337 5340      JMP ,+1
2340 7402      HLT      /WAS INT, INH.
2341 7300      CLA CLL
2342 1115      TAD K5200
2343 6005      RTF
2344 7240      CLA CMA      /CHECK FOR JAM ON GTF
2345 6004      GTF      /GET LINK, ION, TTY FLAG
2346 7041      CIA
2347 1115      TAD K5200      /EXPECTED BITS
2350 7440      SEA
2351 7402      HLT      /WAS LINK, ION, TTY FLAG SET
2352 7300      CLA CLL
2353 6005      RTF      /REPLACE ION, INT INH
2354 7300      CLA CLL
2355 6004      GTF
2356 7041      CIA
2357 1116      TAD K1200
2360 7440      SEA
2361 7402      HLT      /TTY FLAG, ION, NO LINK
2362 5363      JMP ,+1
2363 7402      HLT      /WAS INT INH
2364 7300      CLA CLL
2365 2027      ISB LOOP      /4096 TIMES
2366 5331      JMP ION1
2367 4555      JMS I XDATER      /GO TO NEXT TEST
2370 4773      JMS I XCON1      /GO TO NEXT TEST
2371 5772      JMP I XRTF1      /GO TO NEXT TEST
2372 2400      XRTF1, RTF1
2373 4000      XCON1, CON1
/
/TEST 06
/TEST DF00 + IF00 FROM SAVE FIELD AFTER PI
/USE RTF TO SET THE FLAG AND GTF TO GET THE FLAG
/CHECK INTERRUPT INHIBIT, DO ALL SAVE
/FIELD COMBINATIONS 0 TO 77,
/

2400
*2400
/

2400 7300      RTF1, CLA CLL
2401 4422      JMS I XTFLG      /SET TTY FLAG
2402 1021      TAD ISZ0
2403 3001      DCA 1

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2484 1020      TAD JMP10
2485 3002      DCA 2
2486 3114      DCA XTOR
2487 1114      XSRTF, TAD XTOR
2410 0005      RTF          /MAKE DF 00 + IF 00
2411 5212      JMP .+1
2412 7402      HLT          /WAS INT INH
2413 7300      CLA CLL
2414 0004      GTF          /GET THE FLAGS
2415 0117      AND K0077
2416 7041      CIA
2417 1114      TAD XTOR      /EXPECTED BITS
2420 7440      SZA
2421 7402      HLT          /WAS DF + IF SET
2422 2027      ISL LOOP      /4096 TIMES
2423 5207      JMP XSRTF
2424 1114      TAD XTOR
2425 1120      TAD K0011
2426 3114      DCA XTOR
2427 2113      ISL XCOUNT
2430 5207      JMP XSRTF      /DO THE REST OF 00 + IF 00
2431 1110      TAD K7770
2432 3113      DCA XCOUNT
2433 5634      JMP I XRTG1
2434 2452      XRTG1, R101
2435 0000      NSTK5, 0
/
2436 7604      LAS          /READ SR 9-11
2437 0044      AND K7
2440 7041      CIA
2441 3031      DCA STKS
2442 5635      JMP I NSTK5
/
/SET TTY FLAG
/
2443 0000      TFLG, 0
2444 7200      CLA
2445 0040      SPF
2446 0041      TSP
2447 5246      JMP .-1
2450 7200      CLA
2451 5643      JMP I TFLG      /EXIT
/
/TEST 09
/TEST PROGRAM INTERRUPT IN EXISTING FIELDS
/USE RTF, GTF, RDF AND RIF FOR CHECK
/CHECK PC, AC, SF AND FLAGS AFTER PI
/IF FAILURE OCCURS CHECK XDATA FOR AC DATA,
/LOC. 0 FIELD 0 FOR CORRECT PC AFTER PI,
/AND IPDF FOR CORRECT DF XX + IF XX,
/PROGRAM SHOULD INTERRUPT INHIBIT TILL JMP I ADRS
/IF PI FAILS TO INTERRUPT HLT IN THAT FIELD
/
2452 7300      R101, CLA CLL
2453 4423      JMS I XSTKS

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```

2454 1120      TAD K0011
2455 3260      DCA IPDF
2456 1132      TAD K0017
2457 3010      DCA 0010
2460 0000      IFDF, 0000      /SET TO CURRENT FIELD UNDER TEST
2461 7300      CLA CLL
2462 1260      TAD IPDF
2463 0005      RTF          /SET FIELDS AND TURN ION
2464 0002      IOF
2465 7300      CLA CLL
2466 2537      ISL I K0000
2467 7000      NOP
2470 1537      TAD I K0000
2471 3136      DCA XDATA
2472 1124      TAD K7402
2473 3041      DCA I K0001      /STORE A HLT IN LOC 1 OF THAT FIELD
2474 1133      TAD K6001
2475 3410      DCA I 0010      /ION FOR THAT FIELD
2476 1130      TAD K1000
2477 3410      DCA I 0010      /TAD FOR THAT FIELD
2500 1124      TAD K7402
2501 3410      DCA I 0010      /HLT FOR FAILURE
2502 1010      TAD 10
2503 1057      TAD K7776
2504 3310      DCA ADRS
2505 1134      TAD JMP10
2506 3001      DCA 0001      /SET LOC 1 FOR RETURN AFTER PI
2507 5710      JMP I .+1      /GO TO THAT FIELD
2510 0000      ADRS, 0000
2511 7041      RET, CIA
2512 1136      TAD XDATA
2513 7440      SZA
2514 7402      HLT          /AC DATA FAILED DURING PI
2515 1000      TAD 0000
2516 7041      CIA
2517 1010      TAD 0010
2520 7440      SZA
2521 7402      HLT          /PC FAILED DURING PI
2522 0214      RDF
2523 0224      RIF
2524 7640      SZA CLA
2525 7402      HLT          /SHOULD BE 0 AFTER PI
2526 0004      GTF
2527 0117      AND K0077
2530 7041      CIA
2531 1260      TAD IPDF
2532 7440      SZA
2533 7402      HLT          /GTF OR RTF OR SF FAILED
2534 1010      TAD 0010
2535 7001      IAC
2536 7640      SZA CLA
2537 5261      JMP IPDF-1
2540 2031      ISL STKS
2541 7410      SKP
2542 5750      JMP I XTRMF

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2543 7300      CLA CLL
2544 1120      TAD K0011
2545 1260      TAD IFDF
2546 3260      DCA IFDF      /SET FOR NEXT FIELD
2547 5256      JMP IFDF -2
2550 1050      XTRMF, TRMF
/

/TEST 15
/TEST TIME SHARE IN FIELD 0.
/ALL HLT, OSR, AND IOT INSTRUCTIONS
/SHOULD TRAP IN USER MODE.
/
*2600
/
2600 7300      T1,      CLA CLL
2601 6007      CAF
2602 6264      CUP
2603 6204      CINT
2604 1021      TAD 1820
2605 3001      DCA 1
2606 1020      TAD JMP10
2607 3002      DCA 2
2608 6007      CAF
2609 6007      SKP
2610 7410      JMP ,      /CAF TRAPED
2611 5212      ION
2612 6001      SKP
2613 7410      JMP ,      /ION TRAPED
2614 5215      KCC
2615 6032      SKP
2616 7410      JMP ,      /KCC TRAPED
2617 5220      IOF
2618 6002      SKP
2619 7410      JMP ,      /IOF TRAPED
2620 5223      GTF
2621 6004      SKP
2622 7410      JMP ,      /GTF TRAPED
2623 5226      /THESE INSTRUCTIONS SHOULD TRAP
2624 6001      T2,      ION
2625 6274      CUP*10      /USER MODE
2626 5232      JMP ,*1
2627 7402      HLT
2628 5233      JMP ,      /HLT DID NOT TRAP
/EXECUTIVE MODE
2629 6254      SINT
2630 5235      JMP ,      /SKIP ON TRAP FLAG
2631 6204      CINT
2632 6254      SINT
2633 6254      /FLAG NOT UP
2634 7410      SKP
2635 5241      JMP ,      /CLEAR TRAP FLAG
2636 7604      LAS
2637 7410      SKP
2638 5244      JMP ,      /SKIP ON TRAP FLAG
2639 5241      /TRAP FLAG STILL SET
2640 7604      /SHOULD NOT TRAP
2641 7410      SKP
2642 5244      JMP ,      /LAS TRAPED IN EXECUTIVE MODE

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2645 6244      RMF
2646 6001      ION
2647 5250      JMP ,*1      /RESTORE USER
/USER MODE
2650 7404      OSR
2651 5251      JMP ,      /GO TO USER
/SHOULD TRAP ON OSR
/EXECUTIVE MODE
2652 6254      SINT
2653 5253      JMP ,      /DID NOT TRAP
/SKIP ON TRAP FLAG
2654 6007      CAF
2655 6254      SINT
2656 7410      SKP
2657 7402      HLT
2658 7404      OSR
2659 7410      SKP
2660 5262      JMP ,      /TRAP FLAG NOT CLEARED
2661 6244      RMF
2662 6001      ION
2663 5266      JMP ,*1      /SHOULD NOT TRAP
/ORS TRAPED IN EXECUTIVE MODE
/RESTORE MODE
/USER MODE
2666 6005      RTF
2667 5267      JMP ,      /GO TN USER
/MAKE THE FLAGS
/RTF FAILED TO TRAP
/EXECUTIVE MODE
2670 6254      SINT
2671 5271      JMP ,      /TRAP FLAG NOT SET
2672 6204      CINT
2673 6254      SINT
2674 7410      SKP
2675 7402      HLT
2676 6004      GTF
2677 7410      SKP
2678 5300      JMP ,      /TRAP FLAG NOT CLEARED
2679 6244      RMF
2680 6001      ION
2681 5304      JMP ,*1      /SHOULD NOT TRAP
/TRAPED IN EXECUTIVE MODE
/RESTORE MODE
/USER MODE
2704 6001      ION
2705 5305      JMP ,      /GO TO USER
/ION DID NOT TRAP
/EXECUTIVE MODE
2706 6254      SINT
2707 5307      JMP ,      /SKIP ON TRAP FLAG
2708 7300      CLA CLL
2709 6004      GTF
2710 0126      AND K0100
2711 7450      SNA
2712 7402      HLT
2713 6204      CINT
2714 6254      SINT
2715 7410      SKP
2716 7402      HLT
2717 6002      IOF
2718 7410      SKP
2719 7410      /FLAG NOT CLEARED
2720 6002      /SHOULD NOT TRAP
2721 7410      SKP
2722 7410

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2723 5323      JMP ,      /IOF TRAPED IN EXECUTIVE MODE
2724 6244      RMF ,      /RESTORE MODE
2725 6801      ION ,
2726 5327      JMP ,+1     /GO TO USER

/USER MODE
/TEST CUF AND CUF+18
2727 7684      LAS
2730 5330      JMP ,      /DID NOT TRAP

/EXECUTIVE MODE
2731 6204      CINT
2732 6244      RMF
2733 6264      CUF
2734 6801      ION
2735 5336      JMP ,+1     /STAY IN EXECUTIVE MODE
2736 7404      OSR
2737 7410      SKP
2740 5340      JMP ,      /CUF DID NOT WORK

/TEST THAT INSTRUCTION ARE INHIBITED WHILE IN USER MODE
2741 6204      CINT
2742 6274      CUF+18     /SET USER
2743 6801      ION
2744 5345      JMP ,+1     /GO TO USER

/USER MODE
2745 7240      CMA CLA     /AC=7777
2746 7684      LAS
2747 5347      JMP ,      /SHOULD CLEAR AC
                          /DID LAS TRAP

/EXECUTIVE MODE
2750 7440      SEA
2751 7402      HLT
2752 6204      CINT
2753 6244      RMF
2754 6801      ION
2755 5356      JMP ,+1     /LAS CHANGED AC

/USER MODE
2756 7200      CLA
2757 7404      OSR
2760 5360      JMP ,      /SHOULD NOT READ SR

/EXECUTIVE MODE
2761 7440      SEA
2762 7402      HLT
2763 6204      CINT
2764 6244      RMF
2765 6801      ION
2766 5367      JMP ,+1     /OSR CHANGED AC

/USER MODE
2767 7240      CLA CMA
2770 7602      HLT CLA     /SHOULD CLA
2771 5371      JMP ,      /DID HLT TRAP

/EXECUTIVE MODE
2772 7440      SEA
2773 7402      HLT
2774 6204      CINT
2775 6803      SRQ
2776 7410      SKP

```

```

2777 7402      HLT
2800 7300      CLA GLL     /INTERRUPT REQUEST
2801 1126      TAD K0100
2802 6805      RTF
                          /ENABLE USER

2803 6801      ION
2804 7800      NOP
2805 5206      JMP ,+1

/USER MODE
2806 6832      KCC
2807 5207      JMP ,      /DID KCC TRAP

/EXECUTIVE MODE
2810 6803      SRQ
                          /IS USER FLAG SET

2811 5210      JMP ,+1
2812 6204      CINT
2813 7300      CLA GLL
2814 1126      TAD K0100
2815 6805      RTF
2816 7300      CLA GLL
2817 6801      ION
2820 5221      JMP ,+1     /ENTER USER

/USER MODE
2821 6804      GTF
2822 5222      JMP ,      /DID GTF TRAP

/EXECUTIVE MODE
2823 8126      AND K0100
2824 7440      SEA
2825 7402      HLT
2826 6803      SRQ
2827 5226      JMP ,+1     /IS USER FLAG SET
2830 6204      CINT
2831 6244      RMF
2832 6801      ION
2833 5234      JMP ,+1

/USER MODE
2834 6804      GTF
2835 5235      JMP ,      /GTF DID NOT TRAP

/EXECUTIVE MODE
2836 6254      SINI
2837 5237      JMP ,      /SKIP ON TRAP FLAG
2840 6204      CINT
2841 6254      SINI
2842 7410      SKP
2843 5243      JMP ,      /FLAG NOT UP
                          /CLEAR TRAP FLAG
                          /SKIP ON TRAP FLAG

2844 6801      ION
2845 7410      SKP
2846 5246      JMP ,      /TRAP FLAG STILL SET
2847 6244      RMF
2850 5251      JMP ,+1     /ION TRAPED IN EXECUTIVE MODE
                          /RESTORE USER
                          /GO TO USER

/USER MODE
2851 6202      CIF
2852 5252      JMP ,      /SHOULD TRAP ON CIF
                          /DID NOT TRAP

/EXECUTIVE MODE
2853 6254      SINI
                          /SKIP ON TRAP FLAG

```

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3054 5254      JMP ,      /DID NOT SKIP
3055 6204      CINT      /CLEAR TRAP FLAG
3056 6254      SINT      /TEST IF CLEARED
3057 7410      SKP
3060 7402      HLT      /TRAP FLAG NOT CLEARED
3061 6202      CIF      /SHOULD NOT TRAP
3062 7410      SKP
3063 5263      JMP ,      /CIF TRAPED IN EXECUTIVE MODE
3064 6244      RMP      /RESTORE MODE
3065 6001      ION
3066 5267      JMP ,+1    /GO TO USER

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3067 6214      /USER MODE
3067 5270      RDP      /READ DATA FIELD
3070 5270      JMP ,      /DID RDP TRAP
3071 6254      /EXECUTIVE MODE
3072 5272      SINT
3073 6204      JMP ,      /TRAP FLAG NOT SET
3073 6204      CINT      /CLEAR TRAP FLAG
3074 6254      SINT      /TEST IF CLEARED
3075 7410      SKP
3076 7402      HLT      /TRAP FLAG NOT CLEARED
3077 6214      RDP      /SHOULD NOT TRAP
3100 7410      SKP
3101 5301      JMP ,      /TRAPED IN EXECUTIVE MODE
3102 6040      /EXECUTIVE MODE
3102 6040      SPF      /FLAG SHOULD WORK
3103 6041      TSP
3104 5303      JMP ,+1    /SHOULD SKP
3105 6003      SRQ
3106 5305      JMP ,+1    /SHOULD SKP
3107 6001      ION
3110 7300      CLA QLL
3111 5311      JMP ,      /DID PI WORK
3112 1126      TAD K0100
3113 6005      RTF
3114 6007      CAF
3115 6001      ION
3116 5317      JMP ,+1
3117 6007      /USER MODE
3117 5320      CAF
3120 5320      JMP ,      /DID CAF TRAP
3121 6003      /EXECUTIVE MODE
3122 7402      SRQ
3123 6007      HLT      /USER FLAG UP
3124 6254      CAF
3125 7410      SINT
3126 7402      SKP
3127 7240      HLT      /FLAG CLEARED
3128 7240      /TEST THAT TTI DOES NOT CHANGE AC
3128 7120      CLA CHA  /AC=7777
3128 7120      STL      /LINK=1

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3131 6274      CUF+10
3132 6001      ION
3133 5334      JMP ,+1
3134 6036      /USER MODE
3135 5335      KRB      /SHOULD NOT ZERO LINK OR SHIFT AC
3136 7040      /EXECUTIVE MODE
3137 7440      CHA
3140 5340      SEA      /AC SHOULD=0000
3141 7420      JMP ,      /AC WAS CHANGED
3142 5342      SNL      /LINK SHOULD EQUAL 1
3143 6254      JMP ,      /LINK WAS CHANGE
3144 5344      SINT      /SKIP ON TRAP FLAG
3145 6204      JMP ,      /TRAP FLAG NOT SET
3146 6244      CINT
3147 6001      RMP
3148 5351      ION
3151 6040      JMP ,+1
3152 6040      /USER MODE
3152 5352      SPF      /FLAG
3153 6041      /EXECUTIVE MODE
3154 7410      JMP ,      /DID SPF TRAP
3155 7402      TSP
3156 6254      SKP
3157 5357      HLT      /TTY FLAG
3160 6204      SINT
3161 6244      JMP ,      /TRAP FLAG NOT SET
3162 6001      CINT      /CLEAR TRAP FLAG
3163 5764      RMP
3164 3200      ION
3164 3200      JMP I ,+1    /GO TO USER
3164 3200      , 177+1
3200      *. 177+1
3200      /USER MODE
3200 6001      ION
3201 5201      JMP ,      /ION DID NOT TRAP
3202 6254      /EXECUTIVE MODE
3203 5203      SINT
3204 6204      JMP ,      /SKIP ON TRAP FLAG
3205 6254      CINT      /TRAP FLAG NOT SET
3206 7410      SINT      /CLEAR TRAP FLAG
3207 7402      SKP      /TEST IF CLEARED
3210 6002      HLT      /FLAG NOT CLEARED
3211 7410      IOF      /SHOULD NOT TRAP
3212 5212      SKP
3213 6244      JMP ,      /IOF TRAPED IN EXECUTIVE MODE
3214 6001      RMP      /RESTORE MODE
3215 5216      ION
3215 5216      JMP ,+1    /GO TO USER
3216 6224      /USER MODE
3216 6224      /TEST CUF AND CUF+10
3216 6224      RIF

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3217 5217      JMP ,          /DID NOT TRAP
          /EXECUTIVE MODE
3220 6204      CINT
3221 6244      RMP
3222 6264      CUF          /STAY IN EXECUTIVE MODE
3223 5224      JMP ,+1
3224 7404      OSR
3225 7410      SKP
3226 5226      JMP ,          /CUF DID NOT WORK
          /EXECUTIVE MODE
3227 7240      CLA CMA
3230 6274      CUF +10      /SET UP USER
3231 6001      ION
3232 5233      JMP ,+1
          /USER MODE
3233 7402      HLT          /SHOULD TRAP
3234 5234      JMP ,          /DID HLT TRAP
          /EXECUTIVE MODE
3235 6203      CDF CIF
3236 6264      CUF          /SETUP FOR EXECUTIVE
3237 6204      CINT        /CLEAR INTERRUPT
3240 6001      ION
3241 5242      JMP ,+1
3242 7604      LAS          /SHOULD NOT TRAP
3243 7410      SKP
3244 5244      JMP ,
3245 7450      SNA          /SR AND AC SHOULD NOT EQUAL ZERO
3246 5246      JMP ,          /LAS WAS INHIBITED

          /TEST HLT AND SKIP
3247 6274      CUF+10      /USER SETUP
3250 6001      ION
3251 5252      JMP ,+1      /GO TO USER
          /USER MODE
3252 7412      SKP HLT      /SHOULD TRAP
3253 5253      JMP ,        /DID NOT TRAP
3254 5254      JMP ,        /SKP DID NOT INDEX PC.
          /EXECUTIVE MODE
3255 6254      SINT          /SKP ON TRAP FLAG
3256 5256      JMP ,
3257 6204      CINT        /CLEAR FLAG
3260 6254      SINT        /IS IT CLEAR
3261 7410      SKP          /YES
3262 5262      JMP ,        /NO-FLAG NO CLEAR

          /LOOP PROGRAM
3263 2266      ISE ,+3      /DO FIRST SECTION 4096
3264 5531      JMP I TIME
3265 7410      SKP
3266 6000      0           /COUNT FOR LOOP
3267 5670      JMP I ,+1
3270 3400      , 177+1

3400      *, 177+1

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          /TEST THAT ALL IOTS TRAP IN USER MODE
3400 7200      CLA
3401 1125      TAD K6000      /BASIC IOT
3402 3207      DCA INST      /SET UP
3403 6274      IOTST, CUF+10 /SET FOR USER
3404 6204      CINT        /CLEAR FLAG
3405 6001      ION
3406 5207      JMP ,+1      /GO TO USER MODE
          /USER MODE
3407 6000      INST, 6000    /IOT THAT FAILED
3410 5210      JMP ,        /IOT DID NOT TRAP
          /EXECUTIVE MODE
3411 6254      SINT          /SKIP ON TRAP FLAG
3412 5212      JMP ,        /TRAP FLAG NOT SET
3413 6204      CINT        /CLEAR FLAG
3414 6254      SINT
3415 7610      SKP CLA
3416 7402      HLT          /FLAG DID NOT CLEAR
3417 2207      ISE INST      /CREATE NEW INSTRUCTION
3420 1207      TAD INST      /TESTED ALL IOT?
3421 0130      AND K1000
3422 7650      SNA CLA
3423 5203      JMP IOTST    /NO -- TEST THE REST

          /TEST THAT ALL (HLT AND OSR) TRAP IN USER MODE
3424 1124      TAD K7402      /BASIC HALT INST
3425 3232      DCA INSTA      /SET UP
3426 6274      HALTA, CUF+10 /SET FOR USER
3427 6204      CINT        /CLEAR FLAG
3430 6001      ION
3431 5232      JMP ,+1      /GO TO USER MODE
          /USER MODE
3432 7406      INSTA, HLT OSR /OPERATE TRAP INST
3433 5233      JMP ,        /DID NOT TRAP
          /EXECUTIVE MODE
3434 7000      NOP          /FOR (HLT,SKP)(OSR,SKP)
3435 6254      SINT          /SKIP ON TRAP FLAG
3436 5236      JMP ,        /TRAP FLAG NOT SET
3437 6204      CINT        /CLEAR FLAG
3440 6254      SINT
3441 7610      SKP CLA
3442 7402      HLT          /FLAG DID NOT CLEAR
3443 1232      TAD INSTA
3444 1123      TAD K0004      /GENERATE ALL GROUPS OF
3445 3232      DCA INSTA      /HALTS AND OSR
3446 1232      TAD INSTA
3447 1122      TAD K0002
3450 7640      SEA CLA
3451 5226      JMP HALTA
3452 6244      RMP
3453 6264      CUF
3454 6001      ION
3455 5256      JMP ,+1
3456 6002      IOF          /SHOULD NOT TRAP

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3457 6254      SINT
3460 7410      SKP
3461 7402      HLT      /TRAP FLAG SET
3462 6040      SPF
3463 6041      TSF      /SHOULD SKP
3464 5263      JMP , -1
3465 6001      ION
3466 7410      SKP
3467 7402      HLT      /DID PI INTERRUPT
3470 7402      HLT      /DID PC INCR.

3471 7300      CLA CLL
3472 6004      GTF
3473 0126      AND K0100
3474 7440      SRA
3475 7402      HLT      /SUF SET
3476 7300      CLA CLL
3477 6007      CAF
3500 6264      CUF
3501 7000      NOP

/
/TEST 16
/TEST TIME SHARE IN EXTENDED MEMORY
/NOW TEST USER MODE TRAP IN ALL EXTENDED FIELDS
/IF TRAP ERROR OCCURS HLT IN THAT FIELD
/USE RTF TO SET USER MODE AND GTF TO GET THE FLAGS
/TEST ALL IOT'S FOR TRAP AND RETURN
/
3502 7300      RIQ2,  CLA CLL
3503 6007      CAF
3504 4423      JMS I XSTKS  /CHECK NO. OF FIELDS PRESENT
3505 1040      TAD KCDF
3506 1045      TAD K10
3507 3335      DCA SRD      /SET OF FOR FIRST FIELD
3510 1041      TAD KCIF
3511 1045      TAD K10
3512 3347      DCA SRI      /SET IF FOR FIRST FIELD
3513 1144      TAD K3577    /GET START OF PROGRAM -1
3514 3010      DCA 10
3515 1145      TAD K7745    /NO. OF INSTRUCTIONS TO TRANSFER
3516 3143      DCA SRCO
3517 7040      CHA
3520 3011      DCA 11      /START AT 0000

3521 1335      TAD SRD      /MAKE FLAGS FOR RETURN CHECK
3522 0111      AND K0070
3523 7010      RAR
3524 7012      RTR
3525 3112      DCA XBAV
3526 1347      TAD SRI
3527 0111      AND K0070
3530 1112      TAD XBAV
3531 1142      TAD K1100
3532 3776      DCA I XFDCON
3533 6201      CDF 00

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3534 1410      TAD I 10
3535 6201      SRD,  CDF 00
3536 3411      DCA I 11      /STORE INSTRUCTIONS
3537 2143      ISB SRCO
3540 5333      JMP SRD-2
3541 1021      TAD 1500
3542 3001      DCA 1      /SET FIELD 0 FOR RETURN
3543 1347      TAD SRI
3544 3002      DCA 2
3545 1020      TAD JMP10
3546 3003      DCA 3
3547 6202      SRI,  CIF 00
3550 5002      JMP 2      /GO TO FIELD UNDER TEST
3551 7300      SRRET, CLA CLL
3552 2031      ISB STKS
3553 7410      SKP
3554 5364      JMP EXITT
3555 1335      TAD SRD      /MORE FIELDS
3556 1045      TAD K10      /GO TO CONTROL
3557 3335      DCA SRD      /SET UP FOR NEXT FIELD
3560 1347      TAD SRI
3561 1045      TAD K10
3562 3347      DCA SRI
3563 5313      JMP STAN
3564 7300      EXITT, CLA CLL  /TEST THIS FIELD
3565 6007      CAF      /TEST DONE GO TO BEGIN
3566 6264      CUF
3567 1151      TAD TTB
3570 3547      DCA I XELL
3571 7004      LAR
3572 7000      SMA CLA
3573 5550      JMP I XBELL
3574 7402      HLT      /TIME SHARE ENABLED
3575 5552      JMP I XTRAP  /AN ERROR CONDITION EXISTS.
3576 3632      XFDCON, FDGON /HIT CONTINUE TRY AGAIN

/
/INSTRUCTIONS TO BE TRANSFERRED TO FIELDS
/
3600
*3600
/
3600 7402      HLT      /SHOULD NOT HLT HERE
3601 7402      HLT      /SHOULD NOT TRAP HERE
3602 7300      FDGO,  CLA CLL
3603 1232      TAD FDGON
3604 6005      RTF
3605 5206      JMP , *1      /SET FOR USER
3606 6000      /USER MODE
3607 5207      IOTX,  IOT
3610 7300      /EXECUTIVE MODE
3611 6004      CLA CLL
3612 7041      GTF
3613 1232      CIA
3614 1232      TAD FDGON  /GET THE FLAGS
3615 1232      /FLAGS THAT SHOULD BE PRESENT

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3614 7640      SZA CLA
3615 7402      HLT
3616 6803      SRQ
3617 9216      JMP , -1
3620 6204      CINT
3621 2206      ISZ IOTX
3622 1206      TAD IOTX
3623 1231      TAD F1000
3624 7640      SZA CLA
3625 9202      JMP F000
3626 6202      CIF
3627 9630      JMP I FRET
3630 3951      FRET, SRRET
3631 1000      F1000, 1000
3632 0000      FDCON, 0000
/
/ CHECK SR0=1 FOR MEMORY EXTENSION ONLY
/
3633 7300      XSR0, CLA GLL
3634 7604      LAS
3635 7700      SMA CLA
3636 9931      JMP I TIME
3637 6007      CAF
3640 1151      TAD TT0
3641 3547      DCA I XELL
3642 9550      JMP I XBELL
/
3643 7300      TRAP, CLA GLL
3644 1153      TAD ATRAP
3645 3001      DCA 1
3646 6274      SUF
3647 6001      ION
3650 9251      JMP , +1
3651 7402      HLT
3652 6254      SINT
3653 7410      SKP
3654 7402      HLT
3655 6264      CUF
3656 6007      CAF
3657 9527      JMP I PLACE
/
/ TEST 07
/ CONFIDENCE CHECK ON ALL EXISTENT FIELDS.
/ MAKE SURE DCA I AND TAD I ARE TO CORRECT STACK.
/ MAKE SURE JUMP IS TO CORRECT STACK.
/ CHECK ALL COMBINATIONS.
/ FIELDS WILL CONTAIN THEIR DF NUMBER IN LOC.0
/
4000
+4000
/
4000 0000      CON1, 0000
4001 7300      CLA GLL
4002 3323      DCA F0NUM
4003 3324      DCA NUMX
4004 1040      TAD KCDF

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4005 3232      DCA CONX
4006 1110      TAD K7770
4007 3327      DCA MSTKS
4010 1110      TAD K7770
4011 3031      DCA STKS
4012 1040      TAD KCDF
4013 3214      DCA , +1
4014 6201      CDF
4015 4307      JMS FILCOR
4016 2031      ISZ STKS
4017 5222      JMP , +3
4020 4252      JMS CONCHK
4021 5227      JMP CON2
4022 1045      TAD K10
4023 1214      TAD F0WRD
4024 3214      DCA F0WRD
4025 2324      ISZ NUMX
4026 9214      JMP F0WRD
/
4027 7300      CON2, CLA GLL
4028 1323      TAD F0NUM
4029 3324      DCA NUMX
4032 6201      CONX, CDF
4033 4307      JMS FILCOR
4034 6203      CDF CIF
4035 4252      JMS CONCHK
4036 7300      CLA GLL
4037 1232      TAD CONX
4040 1045      TAD K10
4041 3232      DCA CONX
4042 2323      ISZ F0NUM
4043 2327      ISZ MSTKS
4044 5227      JMP CON2
4045 6203      CDF CIF
4046 6007      CAF
4047 2027      ISZ LOOP
4050 9201      JMP CON1 +1
4051 9600      JMP I CON1
/
4052 0000      CONCHK, 0000
4053 7300      CLA GLL
4054 3324      DCA NUMX
4055 7604      LAS
4056 0044      AND K7
4057 7040      CMA
4060 3031      DCA STKS
4061 1041      TAD KCIF
4062 3263      DCA , +1
4063 6202      CIF
4064 9541      JMP I K0001
4065 7041      RETADD, CIA
4066 1324      TAD NUMX
4067 7450      SNA
4070 9276      JMP , +6
4071 3112      DCA XSAV

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4072 1263      TAD CONCH
4073 0111      AND K0070
4074 1112      TAD XSAV      /INCORRECT STACK REFERENCED,
4075 7402      HLT          /AC BITS 6-8 GOOD FIELD,
4076 7300      CLA CLL       /AC BITS 9-11 BAD FIELD.
4077 2031      ISR STKS
4100 7410      SKP          /CHECK ALL AVAILABLE STACKS.
4101 5652      JMP I CONCHK
4102 1263      TAD CONCH
4103 1045      TAD K10
4104 3263      DCA CONCH     /UPDATE FIELD CHANGE
4105 2324      ISR NUMX
4106 5263      JMP CONCH

4107 0000      /FILCOR, 0000      /INSTRUCTIONS FOR FIELDS
4110 1324      TAD NUMX      /MODIFIED TO DF#
4111 3537      DCA I K0000
4112 1130      TAD K1000
4113 3541      DCA I K0001
4114 1041      TAD KCIF
4115 3522      DCA I K0002
4116 1326      TAD JMPRET
4117 3540      DCA I K0003
4120 1325      TAD XRETAD
4121 3523      DCA I K0004
4122 5707      JMP I FILCOR

4123 0000      FDNUM, 0000
4124 0000      NUMX, 0000
4125 4065      XRETAD, RETADD
4126 5404      JMPRET, JMP I 4
4127 0000      MSTKS, 0000
          S
    
```

```

0000 11110000 00000000 11111111 11111111 11111111 11111111 11111111 11111111
0100 11111111 11111111 11111111 11111111 11111111 11111111 00000000 00000000

0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111110

0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0500 11111111 11111111 11111111 11111111 11111111 00000000 00000000 00000000

0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111110 00000000 00000000

1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111110 00000000 00000000 00000000 00000000

1200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110000

1400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1500 11111111 11111111 11111111 11111111 11111111 11111111 11111110 00000000

1600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1700 11111111 11111111 11111111 11111111 11000000 00000000 00000000 00000000

2000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2100 11111111 11111111 11111110 00000000 00000000 00000000 00000000 00000000

2200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110000

2400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2500 11111111 11111111 11111111 11111111 11111111 10000000 00000000 00000000

2600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

3000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3100 11111111 11111111 11111111 11111111 11111111 11111111 11110000 00000000

3200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10000000
3300 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000

3400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111110

3600 11111111 11111111 11111111 11111111 11111111 11111111 00000000 00000000
3700 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
    
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```

4000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4100 11111111 11111111 11111111 00000000 00000000 00000000 00000000 00000000

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

```

```

ADRS 2510 DONE0 2206 K0004 0123 KNTR 0102
AGAIN1 0711 ENTER 1200 K0011 0120 KRB 0036
AGAIN2 1005 EXFD 1316 K0017 0132 KRTN 0100
ALL0 2244 EXFLO 1302 K0070 0111 KXFLO 0100
ATRAP 0153 EXITT 3504 K0077 0117 LBT 1515
BEGIN 0203 F1000 3631 K0100 0126 LBTSTC 1675
BEGIN1 0200 FCO 0154 K1 0043 LOOP 0027
BELL 1555 FDCON 3632 K10 0045 MDFSMB 1740
CAA 0653 FDGO 3602 K1000 0130 MGTF 2276
CAB 0754 FDNUM 4123 K1100 0142 MIFSMB 1742
CAC 0755 FDWRD 4014 K1200 0116 MOVE 1463
CAD 0741 FILCOR 4107 K20 0103 MQL 7421
CAE 1036 FILOX 1516 K3577 0144 MSTK3 4127
CAF 0007 FILL 2076 K5200 0115 N1 2043
CAG 1420 FRET 3630 K6000 0125 N2 2042
CAI 1133 GOTO0 1510 K6001 0133 NDF 0030
CAX 2263 GTF 6004 K7 0044 NEWDF 1440
CDF 0201 GTF1 2271 K7000 0047 NOFLO 0034
CDF0S 2245 HALTA 3426 K7402 0124 NOMEM 2200
CHDF 1074 HLTS 0671 K7700 0121 NOSTAK 0033
CHECK 2112 IB0 0345 K7707 0050 NSTK3 2435
CIF 0202 IB1 0354 K7717 0056 NUHX 4124
CIFCK 0753 IB2 0402 K7727 0055 OK1 0231
CIFCK1 1046 IB3 0421 K7737 0054 OK2 0257
CIFJMP 0723 IB4 0444 K7744 0071 OK3 0305
CIFJMS 1017 IB5 0463 K7745 0145 OK4 0333
CIFJPL 0715 IB6 0506 K7747 0053 OK5 0373
CIFJSL 1011 IB7 0525 K7757 0052 OK6 0440
CINT 0204 IBSF 0656 K7766 0067 OK7 0502
CKPC 1226 IBSF1 1000 K7767 0051 OK8 0544
CNSTK 2231 IFGN 1605 K7770 0110 P 2044
CON1 4000 IFDF 2400 K7771 0064 PLACE 0127
CON2 4027 INBT 3407 K7772 0063 POINT 0069
CONCH 4063 INSTA 3432 K7773 0062 POS 2237
CONCHK 4052 INTE 1603 K7774 0061 RANA 2057
CONX 4032 INTEP 1674 K7775 0060 RDF 6214
CUF 0204 IOF 0002 K7776 0057 REPEAT 1733
DAT 0032 ION 0001 K7777 0046 RET 2511
DATER 2047 ION1 2331 K78 0066 RETADD 4060
DCAI 0601 IOT 0000 KCAI 0036 RIB 6234
DF0 0211 IOTST 3403 KCAIM 0035 RIF 6224
DF1 0235 IOTX 3606 KCC 0032 RIG1 2452
DF2 0246 IS00 0021 KCDF 0040 RIG2 3502
DF3 0263 JMP2 0104 KCDF1 0156 RMF 6244
DF4 0274 JMP10 0020 KCIF 0041 RMFCN1 1676
DF5 0311 JMP14 1702 KDATA 0157 RMFDY 1703
DF6 0322 JMP1R 0134 KDFSMB 1737 RMFDY1 1710
DF7 0220 JMPRET 4126 KFLD0 0105 RMFE1 1665
DFCN 1677 K0000 0107 KHLT 0037 RMFE2 1656
DFLO 0607 K0001 0101 KIFSMB 1741 RMFI1 1660
DFN 1446 K0002 0102 KJMP 0101 RMFI2 1661
DOAUTO 1317 K0003 0100 KNOP 0752 RMFL1 1633

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RMFL2	1621	XMEM	1781
RMFL3	1615	XNOM	2236
RMFTST	1600	XRANB	0025
RTF	6005	XRET	0135
RTF1	2400	XRETAD	4125
RTRN	1427	XRIG1	2424
SFIB	1400	XRMF	0024
SJNT	6254	XRTF1	2372
SKON	6000	XSAV	0112
SPP	6040	XSOF	2304
SRCO	0143	XSR0	3633
SRD	3535	XSRTF	2407
SRI	3547	XSTKS	0023
SRQ	6003	XTDF	0076
SRRET	3551	XTDF1	0077
STAN	3513	XTFLG	0022
STD	1127	XTOR	0114
STKS	0031	XTRAP	0102
STRMF	1107	XTRMF	2550
SUP	6274	XXSR0	0146
T1	2600		
T2	2627		
TADI	0622		
TAUTO	1432		
TFLD	0630		
TFLG	2443		
TIME	0131		
TRANS	1321		
TRAP	3643		
TRPLD	1337		
TRMF	1050		
TSP	6041		
TTB	0151		
XAUTO	0026		
XBELL	0150		
XCON1	2373		
XCOUNT	0113		
XDATA	0136		
XDATA	0155		
XELL	0147		
XFD	0042		
XFDCON	3576		
XFER	2000		
XFERC1	2046		
XFERC2	2045		
XFERIN	2032		
XFERL1	2030		
XFERL2	2017		
XFERP	1700		
XFIB	0107		
XGTF1	1047		
XION1	2330		

digitalMAINDEC 08-DHMCA-A
CHANGE ORDERORIGINATOR Bill Heavey
TEL EXT 3621 DATE 6/23/75
DISC PROJ NO. V1806806
COST CENTER NO. 301
*MC 1032*MCO NO. 08-DHMCA-00000
SHEET OF
DATE RECEIVED 6-24-75
FIRST ISSUE 6-26-75
FINAL ISSUE 6-27-75**PROBLEM** Inclusion of statements in restriction section (7.) of document to indicate to operator to remove any ROM's that may be present in system before running PDP-8E memory extension and time share control test.

PGM TO BE CHANGED

MAINDEC-08-DHMCA-A

DISP CODE

OPTIONS AFFECTED

KM8-E

CORRECTION

UNDER RESTRICTIONS (7.) include the following:

Memory available must be sequential 4K memory banks.

All ROM's should be removed prior to running this diagnostic.

BREAK-IN/EFFECTIVITYPRODUCT LINES
AFFECTED

18

ITEM NO.	DOCUMENT/PART NO.	OLD REV	NEW REV	DISP CODE	DESCRIPTION OF CHANGE	
1	MAINDEC-08-DHMCA	A			DOCUMENTATION CHANGE; refer to correction.	DOCUMENTATION AFFECTED ☒ DIAGNOSTICS ☐ TECH MANUAL ☐ TESTER ☐ TEST PROG ☐ ENG SPEC ☐ PURCH SPEC FIELD SERVICE AFFECTED ☒ YES ☐ NO ☐ Customer Charge ☐ Product Line Charge ☐ PL _____ ☐ Contact Software Distribution Center for price. ☐ Information Only

DISPOSITION CODES00 - (DELETED)
01 - (DELETED)
02 - USE PRESENT STOCK UNTIL NEW STOCK AVAILABLE (PHASE IN)
03 - REWORK IMMEDIATELY (RETROFIT)
04 - (DELETED)
05 - (DELETED)
06 - DOCUMENT CORRECTION
07 - NEW ITEM (THIS ASSEMBLY)
08 - (DELETED)
09 - SCRAP IMMEDIATELY

*DOL 8/E
DFS 8 FAM
2250
AKK
6-24-75***APPROVAL SIGNATURES**

	Typewritten	Hand Signature
DIAGNOSTIC ENGR.	William Heavey	<i>William Heavey</i>
MFG, ENGR.	Bill Kochman	<i>Bill Kochman</i>
FIELD SERVICE	Carl Cline	<i>Carl Cline</i>
PRODUCT ENGR.	John Kirk	<i>John Kirk</i>

